

AI-Driven Insights for Individualized Learning: Exploring the Effectiveness of Adaptive Systems and Learner Analytic

Name: Mehrunnisa Salim Shaikh

Student ID: 33114790

School of Computing and Engineering University of West London

I hereby declare that the dissertation is all my own work, except as indicated in the text:

Signature: Mehrunnisa Salim Shaikh

Date: 25-09-2024

The copyright of this dissertation belongs to the author under the terms of the UK Copyright acts as amended by The University of West London regulations. Due acknowledgement must always be made of the use of any materials contained in, or derived from this thesis.

Table of Contents:

1. List of figures 3
2. Abstract 5
3. Introduction 5
4. Aims and Objectives 5
5. Background Research of Study 8
6. Literature work 9
    a) Related Work 9
    b) Comparison with Analysis and Evaluation 9
    c) Literature Review 11
7. Experimental Timeline 12
    a) Learner Analytics and Demographics 13
    b) Differences in Learner Analytics 13
    c) Predictive Analysis using learning analytics 15
8. Problem Statement 16
9. Research Questions 16
10. Data and Analysis Method 18
    a) Data Preparation 19
    b) Data Dictionary 19
    c) Columns from the Data 19
    d) Feature Scaling 21
    e) Correlation Analysis 21
    f) Data Pre-processing and Splitting of the Data 22
    g) Model Training 23
        - Algorithms used 24
        - Comparisons of the Algorithms 25
        - Hyperparameter Tuning 26
11. Analytics Tools 26
12. Research Question 1 30

13. Research Question 2 32
    a) The Impact of Age on Learner Analytics: Implications for Academic Achievement
    b) Computation of the Average Age at Which the Graduates and Non-Graduates Join School.
    c) Displaying the Age Demographic of the Graduates and Below Average Students
14. Research Question 3 38
15. Research Question 4 40
16. Results
    a) Decisions from the results 42
    b) Results 44
17. Strategies for Enhancing Student Learning 50
18. Conclusion 51
19. Limitations 52
20. Future Work 53
21. Reference List 53
22. Code Details and Snippets 55

List of Figures:

Figure 1: Dataset columns and rows
Figure 2: Dataset schema
Figure 3: Correlation matrix of the Dataset features
Figure 4: Average academic performance by age group
Figure 5: Droupout and Graduation rates across different age groups
Figure 6: Droupout and Graduation rates across different age groups
Figure 7: Average academic performance by gender
Figure 8: Gender disparities in dropout and graduation rates
Figure 9: Gender distribution among graduates and dropouts
Figure 10: Distribution of age at enrolment for both

Figure 11: Bar Plot on distribution based on Parents Qualification and Occupation
Figure 12: Average academic performance by gender
Figure 13: Plot graph comparison Learner Analytics tools based on Gender
Figure 14: Bar Graph comparison Learner Analytics tools based on Gender
Figure 15: Effectiveness of Learner Analytics in Raising Academic Achievement
Figure 16: Results from the code
Figure 17: Results from the code after hyperparameter tuning
Figure 18: Economical factors and co-relation with student dropouts

Abstract

This research explores the potential of learner analytics to revolutionize education. By collecting and analysing data on student performance, engagement, and behaviour, educators can gain valuable insights into the learning process. This information can be used to inform instructional strategies, identify areas for improvement, and provide targeted support to students. This field has grown significantly since its inception in 2011, and it encompasses both academic research and commercial applications. Learner analytics sits at the intersection of learning sciences, analytics, and human-cantered design. By combining insights from these areas, educators can gain a deeper understanding of student learning and create more effective learning experiences. When we apply the knowledge given to us through real- world exemplifications and case studies, we aim to give preceptors and institutions with practical guidance on employing the power of learner analytics We can analyse easily the reasons and the backends of the students reasoning to dropout, with any additional family responsibilities and that the students might have that will lead to all this and with the showcasing of common factors and details. It involves using criteria and information to gain perceptivity that inform better courses of action. Understanding the requirements of scholars is veritably important from every aspect, there can be colourful reasons as to why a particular pupil would be affected in his or her studies.

Introduction

The use of data-driven decision-making and the incorporation of technology into pedagogy have recently undergone a profound transformation in the field of education. The development of "learner analytics," which involves the systematic gathering, analysis, and application of data to improve educational experiences for students, has sparked this change. We can analyse easily the reasons and the backends of the students reasoning to dropout, with any additional family responsibilities and that the students might have that will lead to all this and with the showcasing of common factors and details. The goal of this study is to add significant knowledge on the practical application of learner analytics in today's many educational contexts. It expands on the growing field of learner analytics. This study intends to close the knowledge gap by analyzing the impact of demographic variables on the efficacy of learner analytics and to provide institutions and educators with evidence-based suggestions for improving student learning experiences in the digital age. . We can assay fluently the reasons and the backends of the scholar’s logic to powerhouse, with any fresh family liabilities and that the scholars might have that will lead to all this and with the showcasing

of common factors and details. The thing of this study is to add significant knowledge on the practical operation of learner analytics in moment's numerous educational surrounds. It expands on the growing field of learner analytics. Understanding the needs of students is very important from every aspect, there can be various reasons as to why a particular student would be affected in his or her studies. The thing of this study is to add significant knowledge on the practical operation of learner analytics in moment's numerous educational surrounds. It expands on the growing field of learner analytics.

Aims and Objective

Aim 1: Age and Academic Success

Investigate the correlation between students' ages at enrolment and their subsequent academic performance and retention rates in higher education.

Aim 2: Gender Biasness in Higher and Further Education

Examine gender-based differences in academic performance, participation, and outcomes, including credit accumulation, enrolment rates and graduation rates and dropouts.

Aim 3: The Impact of Demographics on Analytics of Learner

Explore how demographic factors, such as age and gender, influence the effectiveness of learner analytics in enhancing student academic achievement in higher education institutions.

Objectives:

Age Objectives:

- Age and Academic Performance: Examine the relationship between students' ages when they start their studies and their academic success, as measured by course credits earned.
- Age and Student Retention: Investigate how age affects student retention rates, including both success and dropout rates.
- Age and Learner Analytics: Explore patterns in the use of learner analytics tools among students of different ages.

Gender Objectives:

- Gender and Academic Performance: Examine how gender affects academic outcomes, including course credits earned, enrollment patterns, and course completion rates.
- Gender and Student Retention: Investigate gender-related differences in student retention and graduation rates.
- Gender and Learner Analytics: Analyze the connection between gender and the use of learner analytics tools.

Demographic Characteristics Objectives:

- Demographic Factors and Learner Analytics: Explore how factors like age and gender impact the adoption and use of learner analytics tools in education.
- Impact on Academic Achievement: Examine the effectiveness of learner analytics in improving academic outcomes for students from various backgrounds.
- Inclusive Learner Analytics: Identify strategies to ensure that learner analytics tools are accessible and responsive to the needs of all students, regardless of their demographic characteristics.

Background Research of the Study

The use of data-driven decision-making and the incorporation of technology into pedagogy have recently undergone a profound transformation in the field of education. The development of "learner analytics," which involves the systematic gathering, analysis, and application of data to improve educational experiences for students, has sparked this change. Education is undergoing a data revolution! Technology advancements have allowed us to gather a wealth of student information through systems like SIS and LMS. This data, analysed through "learner analytics," holds immense promise for improving how we teach and learn.

However, despite the abundance of data, there's a gap between insights and practical applications. This research aims to bridge that gap by focusing on how learner analytics can be effectively used in various educational settings. We'll be looking at how learner demographics might influence the effectiveness of these analytics. The ultimate goal is to provide educators and institutions with data-driven strategies for enhancing student learning experiences in this digital age. In simpler terms, we're moving beyond guesswork and using data to make informed decisions about education! This research will contribute to the growing field of learner analytics and offer practical guidance for educators and institutions. There are still difficulties in turning data insights into tactics that educators and institutions can use, despite the abundance of data and the great promise of learner analytics. The educational landscape is rapidly transforming, with data playing an increasingly important role. This study aims to bridge a crucial knowledge gap by examining how learner demographics influence the effectiveness of learner analytics. By analysing this data, the research hopes to provide educators and institutions with concrete, evidence-based recommendations. This will ultimately lead to improved learning experiences for students in the digital age, ensuring they are effectively supported on their educational journeys. The goal of this study is to add significant knowledge on the practical application of learner analytics in today's many educational contexts. It expands on the growing field of learner analytics. This research aims to bridge the knowledge gap by examining how demographic factors influence the effectiveness of learner analytics. By providing institutions and educators with data-driven insights, we hope to enhance student learning experiences in today's digital age. Students can be impacted by a wide range of challenges that lead them to leave school before completing their degree. These reasons can be academic, financial, personal, or a combination of these factors.

Literature Work

Related Work:

Comparison with Analysis and Evaluation

YEAR	AUTHOR(S)	OBJECTIVE	DM TECHNIQUE	LIMITATIONS
2007	Ogor	Development of a methodology through performance prediction for deployment of a student performance assessment & monitoring system	C5.0	Lack the integration of students' attitude and so on to the result of mining student performance.
2008	Romero, Ventura, Espejo & Hervas	Comparison of different DM methods and techniques for classifying students	CART, GAP, GGP, AprioriC and so on.	The Moodle DM tool developed does not give an overall generalization of prediction of SAP for all forms of learning.
2009	Dekker, Pechenizkiy & Vleeshouwer	Generation of predictive models form selected DM techniques to predict Electrical Engineering students' dropouts	CART, C4.5, BayesNet, Logistic model, JRip & RF.	Students' progression was not taken into consideration in the study.
2010	Mohsin, Norwawi, Hibadullahi & AbdWahab	Investigation of factors influencing the final programming performance	Rough Set	Prominent DM techniques like NB, SVM need to be applied on the dataset with other attributes added to confirm results.
2011	Yadav, Bharadwaj & Pal	Development of DT-based models to predict students' performance.	CART, ID3 & C4.5	Only one evaluation criteria were used and that is accuracy which is not enough to conclude.
2012	Yadav and Pal	Performance improvement of Engineering students using DM	C4.5, ID3, CART	Only DT techniques were used, other prominent DM techniques were not considered.
2012	Pal	Development of a DM application to generate predictive models for Engineering student's dropout management	NB	The variables used in the study do not include any social behaviour feature.
2012	Yildiz *et al.*	Using model to evaluate the Distance Education Students' academic performances.	Genetic-Fuzzy hybrid.	Only meta-heuristic algorithms used, no focus on any of the traditional DM algorithms.
2013	Do & Chen	Presentation of a Neuro-fuzzy approach to classification of SAP	Neuro-Fuzzy	The approach being fuzzy based can be rigid with rigid fuzzy rule sets.
2013	Aziz, Ismail & Ahmad	Development of a prototype SAP prediction model	NB	The study did not take the student's social behaviour into consideration.

2014	Ogunde & Ajibade	Development of a DM system to predict student's graduation grades based on their entry results data using ID3	ID3	More distinctive attributes like social behaviour of students that can have a direct impact on the final prediction needs to be considered for better accuracy.
2014	Chen, Hsieh & Do	To train Neural Network with two meta-heuristic algorithms of CS and COA for SAP prediction.	NN trained with CS & COA	It only serves as an effective standard for NN and not for other DM algorithms.
2015	Aziz, Ismail, Ahmad & Hassan	Development of a framework for predicting the performance of 1st year Bachelor students in Computer Science	NB	Social behaviour of students was not considered in the study.
2015	Joseph & Devadas	Development of a DT-based classification model to predict SAP	Weighted modified ID3	The developed model was evaluated using only the prediction accuracy as the performance metric.
2015	Lakkaraju *et al.*	Development of a machine learning framework to identify students at risk	RF, AB, LR, SVM, DT	The study needs to employ other commonly used EDM techniques like NB for generalization of results.
2015	Kaur, Singh & Josan	Prediction of slow learners	MLP, NB, SMO, J48 & REPTree	Some other important factors like social behaviour of students are not considered in the study.
2015	Strecht, Cruz, Soares, Mendes-Moreira & Abreu	Evaluation of the most popular classification and regression algorithms for predicting SAP	SVM, CART, NB, RF, kNN, AB, AB.R2, OLS	The evaluation criteria used in the study is statistically inclined and is not goal oriented performance metrics used in data mining.
...				
2016	Aulck, Velagapudi, Blumenstock & West	Predicting Students dropout	Regularised LR, kNN and RF	The DM methods employed are not the most popular in EDM researches and may limit research result acceptance.
2016	Al-Barrak & Al-Razgan	Prediction of student's GPA based on grades in previous courses	J48	The focus is mainly on the grades in courses not on other influencing factors on GPA
2017	Asif, Merceron, Ali & Haider	To analyze the performance of undergraduate students using EDM	DT, NN, NB, RI-IG, k-NN, X-means with pragmatic policy	The study took student progression into consideration but there is a need for generalizability of the results of DM.
2017	Asif, Hina & Haque	Studying DM techniques to predict the students' graduation performance in the final year.	DT-GI, DT-IG, DT-ACC, NB, NN, RF-GI, RF-IG, RF-ACC	Some other important DM techniques were not studied apart from the DT variants.
2017	Gilbert	Prediction of student outcomes using RF and genetic algorithm for feature selection and pruning	Random Forest (RF)	The DM algorithm employed is not a common one in the area and therefore the obtained result may not be popular.

2018	Yang, Lu, Huang, Huang, Ogata & Lin	Improving Multi LR predictive capability with PCA	Multiple Linear Regression, PCA	Prediction model developed is strictly associated with the learning activity and particular data in the study thereby making the model non-flexible.
2018	Adejo & Connolly	Predicting SAP using a multi-model heterogeneous ensemble approach	ANN, SVM, DT, Ensemble	Non-generalizable models.
2018	Bendangnuksung & Prabu	Using Deep Neural Network to predict students' performances.	Deep NN	The study can be improved by adding some other DM techniques.
2018	Razaque *et al.*	Using NB to perform SAP prediction	NB	Non-generalizable models.
2019	Hasheminejad & Sarvmili	Performing SAP prediction using PSO	PSO	There is need for inclusion of any of the traditional DM techniques.
2019	Burman & Som	Investing RBF and Linear kernels for SVM prediction of SAP	SVM	Focus is only on finding the best among the two kernels of SVM
2019	Afeni, Oloyede & Okurinboye	Predicting Students' Performance using classification Algorithm	ID3 & C4.5	Comparison of classifiers is only among Decision Trees. There is need to extend to other DM techniques.
2019	Chen *et al.*	Developing a hybrid of DT & ELM for dropout prediction	DT & ELM	Can be used to provoke hybrid of two or more other DM techniques
2019	Francis & Babu	Predicting SAP using Hybrid	SVM, NB, DT & NN with K-means algorithm.	The study only finds the best based on selected group in dataset

Literature Review:

A study by Fabelo et al. (2011) examines the financial burden of strict disciplinary actions in Texas schools, such as suspensions and expulsions. Their findings reveal a troubling link: harsher punishments increase the likelihood of students repeating grades or dropping out altogether. This has a significant economic impact. Students who fall behind or leave school early typically earn less throughout their lives, leading to a loss of potential tax revenue for the state. The study estimates this loss at roughly $68 million annually. Furthermore, the research highlights a racial disparity in these punishments, with minority students disproportionately affected. This translates to a more substantial financial burden – an estimated $711 million lost each year due to increased dropout rates among minority students. To address these issues and associated costs, the researchers

advocate for alternative behaviour management strategies and improved support systems for struggling students.

In addition, the study offered a practical tool (a linear model) to pinpoint strategies that universities can use to improve student achievement. Focusing on impact: The study went a step further by proposing a linear model that can be used to develop more effective strategies for improving university student outcomes.

Ali et al. (2013) investigated how factors like gender, age, academic program, school background, socioeconomic background, and daily study time influence the academic performance of graduate students at the Islamia University of Bahawalpur's Rahim Yar Khan Campus. They used surveys and statistical methods like linear regression and correlation to analyse data from students. Their findings revealed that age, socioeconomic background, and daily study hours significantly affect academic achievement. The study even proposed a linear model to develop strategies for improving university students' performance.

Experimental Timeline

- **Experimental Timeline and Iterations**

The analysis would likely be iterative, with each step informing the next:

- **Initial Exploration**: Identify patterns and correlations.
- **Hypothesis Testing**: Conduct statistical tests.
- **Modeling**: Build and validate predictive models.
- **Reporting**: Interpret and present the findings.

Research has consistently shown that learner analytics can positively influence student academic achievement. By providing teachers with valuable information about student progress, analytics can enable timely interventions to address knowledge gaps and enhance overall performance. Johnson et al. (2018) underscored the importance of data-informed decision-making in higher education, emphasizing learner analytics as a key factor in improving student outcomes.

West et al.'s (2019) study focused on the impact of learner analytics on student retention. Their findings revealed a significant increase in retention rates among students who received targeted

interventions based on analytics data. This demonstrates that learner analytics not only helps identify at-risk students but also supports strategies to ensure their continued academic success.

It's crucial to note that the benefits of learner analytics may not be evenly distributed across all demographic groups. Empirical evidence suggests that age can significantly influence how students engage with and benefit from analytics tools. Studies by Young et al. (2019) and Kim and Lee (2020) indicate that older students may face challenges in adapting to and fully utilizing analytics tools.

Gender-based disparities in learner analytics usage have also been explored. Johnson and Smith (2018) found that female students were more likely than male students to use analytics dashboards to monitor their progress and seek assistance, which may positively impact their performance.

Learner Analytics and Demographics

The learner analytics tool promises to improve all the students performances through data driven insights for a change and can be used as a powerful engine for education. The potential that it significantly increases the performance of the students by checking their work and by these demographics which gives us an idea as to how complex the relationship between the learner analytics and students outcomes can be in one of the significant studies, Anderson and Dron (2019). Thoroughly investigated how these learner analysis on people learning can be derived there results not only highlighted the beneficial relationship between the use of these tools, but it also showed how it improved the students performances, learner analytics also gives us the ability to pinpoint areas for development as we can give the students a feedback and interventions which can help them raise their grades and improve their significant and satisfaction for the learning processes

Differences in Learner Analytics:

There is a research study done by Smith and Johnson‘s (2019) on gender differences in which they use and show the learner analytics dashboard, which is quite interesting to understand because according to the study, the female students were shown to be more likely active in using these analytical tools than the male students to track their academic progresses. The idea around this is that the female students may demonstrate more data which is used in informing dates,

decision-making behaviours throughout the study and learning journey and it is consistent and proactive approach is helpful in monitoring the performance and seeking guidance.

Elsewhere, if we observe Kim and leave 2020 research paper which says that when they focus on gender when they look at factors influencing London analytics their adoption in higher education. For this system in general, the students were found to be more open to using these analytical tools than the female students according to their research, but the complexity of these gender based disparities or bias, learner, analytics says and highlights that despite the contradiction with the earlier studies, it's still implies the different locations and student populations may have different attitudes and their preferences can be different towards gender.

But that does not tend us to ignore the fact that gender cannot be taken into consideration to find an analyse data along with the other characteristics like learning preferences and style.

Ethical issues with learner analytics

A new era of data-driven decision-making has begun with the incorporation of learner analytics into education, but it has also raised a number of ethical questions that need careful examination. Data security and privacy are still of utmost importance because unauthorized access or data breaches may expose students' private data. Recent high-profile data breaches in educational institutions have brought these vulnerabilities to light (Noble, *et al.* 2018). An important ethical consideration is obtaining the students' informed agreement for data collection and use. Concerns about fairness and bias are still present. Learner analytics systems may add bias to decision-making if they are not carefully built, which could exacerbate disparities (Lemov, *et al.* 2018). Continuous monitoring, transparency, and algorithmic fairness are essential for mitigating bias (Selwyn, *et al.* 2019). Trust must be upheld inside educational institutions through transparency and accountability. Data collection, analysis, and usage procedures should be made available to students and other interested parties (Lauder, *et al.*2018). It is necessary to set up accountability systems, such as distinct channels for voicing complaints (Sclater, *et al.*2018). Ownership and access to data have also been questioned. There are many institutions, software companies, and outside suppliers involved in determining who owns the data and how it can be accessed or shared (Knox, *et al.*2019). To manage data responsibly, it is crucial to establish explicit policies and agreements. Since learner analytics may result in the classification of students based on their academic performance and habits, addressing concerns about profiling and stigmatization is crucial. The overall educational experiences of pupils as well as their self-esteem may suffer as a result (Lemov, *et al.*2018).

Predictive Analysis using learning analytics

The provided research paper that delves into the realms of Educational Data Mining (EDM) and Learning Analytics (LA) with a specific focus on predicting student success, particularly in the context of higher education institutions in Pakistan. (Ali Daud, Naif Radi Aljohani, Rabeeh Ayaz Abbasi, Miltiadis D. Lytras, Farhat Abbas, and Jalal S. Alowibdi.)

The abstract begins by emphasizing the significance of forecasting student performance as it directly impacts a country's social and economic growth by producing skilled graduates. It notes that existing methods mainly rely on features related to academic performance, family income, and assets while largely ignoring factors such as family expenditures and students' personal information. The primary objective of this research is to explore these often-overlooked feature sets by collecting data from scholarship-holding students across various Pakistani universities. To achieve this, the study employs learning analytics and classification models, both discriminative and generative, to predict whether a student will complete their degree or not. The abstract suggests that the proposed methodology significantly outperforms existing approaches, primarily due to its incorporation of family expenditures and personal information as influential factors. It concludes by highlighting the potential impact of the research on improving policies in higher education and, in essence, underscores the importance of considering a broader range of features in predicting student success.

Learning analytics involves the collection and analysis of data to predict and improve student success, primarily at the course level. The text highlights various goals, types of data, and benefits associated with learning analytics.(Beth Dietz-Uhler & Janet E. Hurn)

One key goal of learning analytics is to enhance student success and retention by providing real-time insights into student performance. It also aims to improve teaching practices by allowing faculty to identify areas where course improvements are needed. Learning analytics relies on data, typically accessible through Learning Management Systems (LMS), which can include information about student engagement, grades, and resource access.

Problem Statement

While learner analytics have become a hot topic in education, promising data-driven ways to improve student learning, a crucial question remains unanswered. How effectively can these analytics be applied to diverse student groups with varying backgrounds? Research in this area is scarce. Age, gender, and socioeconomic status can all influence learning styles and needs. It's unclear if learner analytics can provide insights that address these differences.

This is a pressing issue because schools are accumulating vast amounts of student data through systems like SIS and LMS. Without understanding how learner analytics translate into solutions that consider diverse demographics, educators and institutions risk widening the achievement gap between student groups. This knowledge gap hinders us from fully utilizing the potential of learner analytics. Concerns about data privacy combined with the ethical ramifications of handling sensitive demographic data provide further difficulties. The path towards data-driven education isn't without its hurdles. While learner analytics hold immense promise, they raise concerns about student data privacy. Furthermore, the ethical implications of handling sensitive demographic information must be carefully considered. These challenges add another layer of complexity to utilizing learner analytics effectively, requiring careful consideration of both the benefits and potential drawbacks. These problems highlight how crucial it is to deal with the situation at hand. In order to determine if learner analytics may be successfully used to improve student learning outcomes in a way that is inclusive, equitable, and morally acceptable, this research seeks to clarify the relationship between demographic characteristics and learner analytics.

Research Questions

Research Q1 -

Research Question: How does the age at which students enroll in a course impact their academic performance and retention rates, as measured by the number of curricular units credited and the target outcome (dropout/graduate)?

Objective: This study aims to investigate the relationship between students' ages at enrollment and their subsequent academic performance and retention. By analyzing the number of credited curricular units and graduation/dropout rates, we will explore how age influences these outcomes.

The Role of Learner Analytics: Through the application of learner analytics, we can collect, analyze, and report data on learners and their contexts. This data-driven approach enables us to

develop educational strategies tailored to specific age groups, thereby enhancing the overall student experience and improving outcomes.

Research Q2 -

Gender-Based Differences in Academic Performance and the Role of Learner Analytics

Research Question: How does gender correlate with academic performance, participation, and outcomes, as measured by the number of curricular units credited, enrolled, and approved?

Objective: This study aims to investigate the relationship between gender and various academic aspects, including performance, participation, and outcomes, as measured by the number of credited, enrolled, and approved curricular units.

The Role of Learner Analytics: Learner analytics can be instrumental in identifying patterns and trends related to gender disparities in academic settings. By leveraging these analytics, institutions can develop gender-sensitive pedagogies and support mechanisms that promote equality and inclusivity.

Research Q3 -

The Impact of Demographics on Learner Analytics: A Case for Inclusivity

Research Question: How do demographic characteristics, such as age, influence the use and effectiveness of learner analytics in enhancing student academic achievement in higher education institutions?

Objective: This study aims to understand how demographic factors, particularly age, affect the utilization and efficacy of learner analytics in improving academic outcomes.

The Role of Learner Analytics: Learner analytics tools must be designed and implemented with consideration for the diverse demographic makeup of the student body. This research will inform the development of more inclusive and effective learning analytics strategies that are responsive to the varied needs and preferences of the student population.

Research Q4 -

Predicting Student Dropout: A Learner Analytics Approach

Research Question How can we use data analysis to accurately identify students who may drop out of school, and what can we do to help them stay in school and succeed?

Objective: This study aims to investigate the application of predictive analytics in accurately predicting student dropout and to develop targeted interventions based on these predictions to enhance learning analytics and reduce dropout rates.

The Role of Learner Analytics: Predictive analytics can leverage historical data and student behavior patterns to identify students at risk of dropping out. By integrating these insights into learning analytics systems, institutions can provide timely support and interventions, ultimately improving student retention and success.

Data and Analysis Method

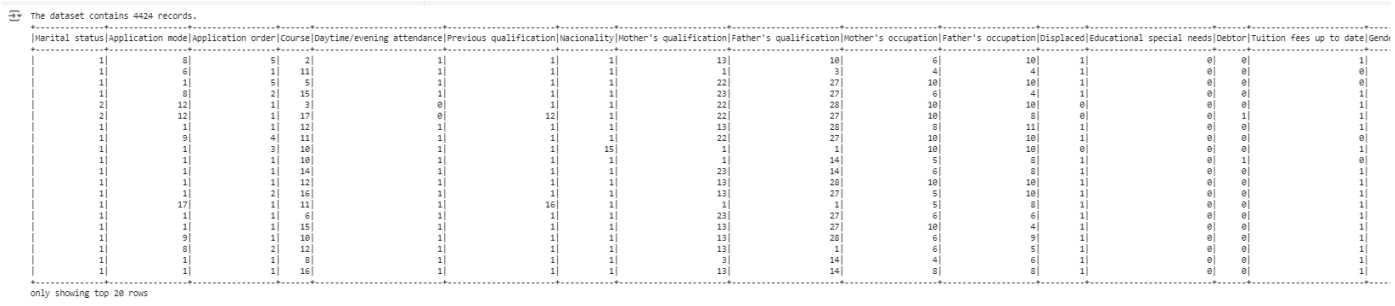
This dataset offers a comprehensive perspective on undergraduate students enrolled in diverse degree programs at a higher education institution. It encompasses a wide range of information, including demographic details, socioeconomic indicators, and academic performance metrics, which can be leveraged to analyze potential factors influencing student dropout rates and academic achievements. The dataset comprises multiple separate databases, each containing relevant information captured at the time of enrolment. This information encompasses application methods, marital status, selected courses, and more.

Furthermore, the dataset enables the assessment of overall student performance at the end of each semester by tracking metrics such as curricular units taken, enrolled courses, grades obtained, and course approvals. To enhance our understanding of the factors impacting student outcomes, the dataset also includes regional economic data, such as the unemployment rate, inflation rate, and GDP. These economic indicators offer insights into how economic conditions may correlate with student dropout rates and academic success across various fields of study, including agronomy, design, education, nursing, journalism, management, social services, and technology.

In essence, this dataset serves as a valuable analytical tool, shedding light on the factors that influence students' decisions to persist in their studies or discontinue them across a diverse array of academic disciplines.

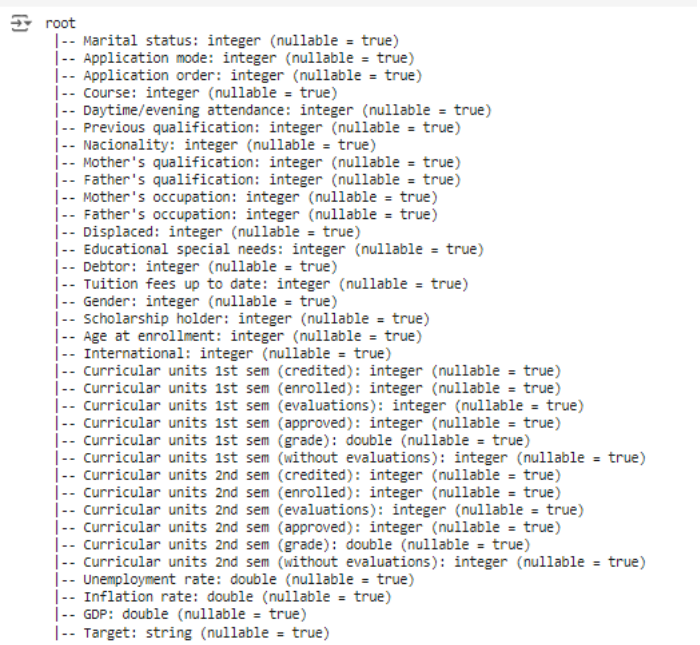
The following link is for the Dataset from Kaggle:
https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention/data

Data Preparation:



```
The dataset contains 4424 records.
+--------------+----------------+-----------------+------+-----------------------------+-----------------------+-----------+---------------------+---------------------+---------------------+---------------------+-----------+-------------------------+-------+-------------------------+------+
|Marital status|Application mode|Application order|Course|Daytime/evening attendance   |Previous qualification |Nacionality|Mother's qualification|Father's qualification|Mother's occupation|Father's occupation|Displaced|Educational special needs|Debtor|Tuition fees up to date|Gende
+--------------+----------------+-----------------+------+-----------------------------+-----------------------+-----------+---------------------+---------------------+---------------------+---------------------+-----------+-------------------------+-------+-------------------------+------+
|             1|               8|                5|     2|                            1|                      1|          1|                   13|                   10|                    6|                   10|          1|                        0|      0|                        1|
|             1|               6|                1|    11|                            1|                      1|          1|                    1|                    3|                    4|                    4|          1|                        0|      0|                        0|
|             1|               1|                5|     5|                            1|                      1|          1|                   22|                   27|                   10|                   10|          1|                        0|      0|                        0|
|             1|               8|                2|    15|                            1|                      1|          1|                   23|                   27|                    6|                    4|          1|                        0|      0|                        1|
|             2|              12|                1|     3|                            0|                      1|          1|                   22|                   28|                   10|                   10|          0|                        0|      0|                        1|
|             2|              12|                1|    17|                            0|                     12|          1|                   22|                   27|                   10|                    8|          0|                        0|      1|                        1|
|             1|               1|                1|    12|                            1|                      1|          1|                   13|                   28|                    8|                   11|          1|                        0|      0|                        1|
|             1|               9|                4|    11|                            1|                      1|          1|                   22|                   27|                   10|                   10|          1|                        0|      0|                        0|
|             1|               1|                3|    10|                            1|                      1|         15|                    1|                    1|                   10|                   10|          0|                        0|      0|                        1|
|             1|               1|                1|    10|                            1|                      1|          1|                    1|                   14|                    5|                    8|          1|                        0|      1|                        0|
|             1|               1|                1|    14|                            1|                      1|          1|                   23|                   14|                    6|                    8|          1|                        0|      0|                        1|
|             1|               1|                1|    12|                            1|                      1|          1|                   13|                   28|                   10|                   10|          1|                        0|      0|                        1|
|             1|               1|                2|    16|                            1|                      1|          1|                   13|                   27|                    5|                   10|          1|                        0|      0|                        1|
|             1|              17|                1|    11|                            1|                     16|          1|                    1|                    1|                    5|                    8|          1|                        0|      0|                        1|
|             1|               1|                1|     6|                            1|                      1|          1|                   23|                   27|                    6|                    6|          1|                        0|      0|                        1|
|             1|               1|                1|    15|                            1|                      1|          1|                   13|                   27|                   10|                    4|          1|                        0|      0|                        1|
|             1|               9|                1|    10|                            1|                      1|          1|                   13|                   28|                    6|                    9|          1|                        0|      0|                        1|
|             1|               8|                2|    12|                            1|                      1|          1|                   13|                    1|                    6|                    5|          1|                        0|      0|                        1|
|             1|               1|                1|     8|                            1|                      1|          1|                    3|                   14|                    4|                    6|          1|                        0|      0|                        1|
|             1|               1|                1|    16|                            1|                      1|          1|                   13|                   14|                    8|                    8|          1|                        0|      0|                        1|
+--------------+----------------+-----------------+------+-----------------------------+-----------------------+-----------+---------------------+---------------------+---------------------+---------------------+-----------+-------------------------+-------+-------------------------+------+
only showing top 20 rows
```

Figure 1: Dataset columns and rows



```
root
 |-- Marital status: integer (nullable = true)
 |-- Application mode: integer (nullable = true)
 |-- Application order: integer (nullable = true)
 |-- Course: integer (nullable = true)
 |-- Daytime/evening attendance: integer (nullable = true)
 |-- Previous qualification: integer (nullable = true)
 |-- Nacionality: integer (nullable = true)
 |-- Mother's qualification: integer (nullable = true)
 |-- Father's qualification: integer (nullable = true)
 |-- Mother's occupation: integer (nullable = true)
 |-- Father's occupation: integer (nullable = true)
 |-- Displaced: integer (nullable = true)
 |-- Educational special needs: integer (nullable = true)
 |-- Debtor: integer (nullable = true)
 |-- Tuition fees up to date: integer (nullable = true)
 |-- Gender: integer (nullable = true)
 |-- Scholarship holder: integer (nullable = true)
 |-- Age at enrollment: integer (nullable = true)
 |-- International: integer (nullable = true)
 |-- Curricular units 1st sem (credited): integer (nullable = true)
 |-- Curricular units 1st sem (enrolled): integer (nullable = true)
 |-- Curricular units 1st sem (evaluations): integer (nullable = true)
 |-- Curricular units 1st sem (approved): integer (nullable = true)
 |-- Curricular units 1st sem (grade): double (nullable = true)
 |-- Curricular units 1st sem (without evaluations): integer (nullable = true)
 |-- Curricular units 2nd sem (credited): integer (nullable = true)
 |-- Curricular units 2nd sem (enrolled): integer (nullable = true)
 |-- Curricular units 2nd sem (evaluations): integer (nullable = true)
 |-- Curricular units 2nd sem (approved): integer (nullable = true)
 |-- Curricular units 2nd sem (grade): double (nullable = true)
 |-- Curricular units 2nd sem (without evaluations): integer (nullable = true)
 |-- Unemployment rate: double (nullable = true)
 |-- Inflation rate: double (nullable = true)
 |-- GDP: double (nullable = true)
 |-- Target: string (nullable = true)
```

Figure 2: Dataset schema

Data Dictionary:

Marital Status:	Indicates the marital status of the student.
Application Mode:	Represents the method used to submit the application (e.g., online, in-person).
Application Order:	Indicates the order in which applications were received or processed.
Course:	Refers to the specific course in which the student is enrolled.
Daytime/Evening Attendance:	Indicates whether the student attends classes during the day or evening.

Previous Qualification:	Refers to the qualifications the student held prior to joining the course.
Nationality:	Represents the nationality of the student.
Mother's Qualification:	Indicates the educational qualifications of the student's mother.
Father's Qualification:	Indicates the educational qualifications of the student's father.
Mother's Occupation:	Refers to the occupation of the student's mother.
Father's Occupation:	Refers to the occupation of the student's father.
Target:	The target variable indicating whether a student graduated or dropped out.
Additional Columns	The dataset likely includes numerous other columns related to student academic performance, such as ● Number of curricular units credited, enrolled, evaluated, and approved in each semester ● Economic indicators like unemployment rate, inflation rate, and GDP

Columns from the Data

Features: The dataset has certain demographic characteristics such as marital status, age at the time of enrolment, nationality, and previous qualifications to name but a few. These features seem likely to be encoded into numbers as well.

Target: The target variable is labelled as Target which classifies a student to be either a Dropout or a Graduate.

Feature Encoding

Binary Encoding: Convert the "Dropout" or "Graduate" target variable into binary values, with "Dropout" represented by 1 and "Graduate" by 0.

Missing Value Handling: Identify and address any missing values in the dataset to ensure data integrity.

Feature and Target Separation: Divide the dataset into features (X) and the target variable (y).

Data Splitting: Partition the data into training and testing sets to evaluate the model's performance on unseen data.

Features	Encoding
Dropout	1
Graduate	0

Feature Selection and co-relation analysis

Feature Selection: A Key Step in Machine Learning

Feature selection is a crucial process in machine learning pipelines. It involves identifying the most relevant features that contribute significantly to a model's predictive power. One effective technique for feature selection is correlation analysis, which examines the relationship between each feature and the target variable.

Correlation Analysis:

Correlation analysis helps understanding the relationship between each feature and the target feature. Showing the independent and dependent variables that are going to affect my target values.

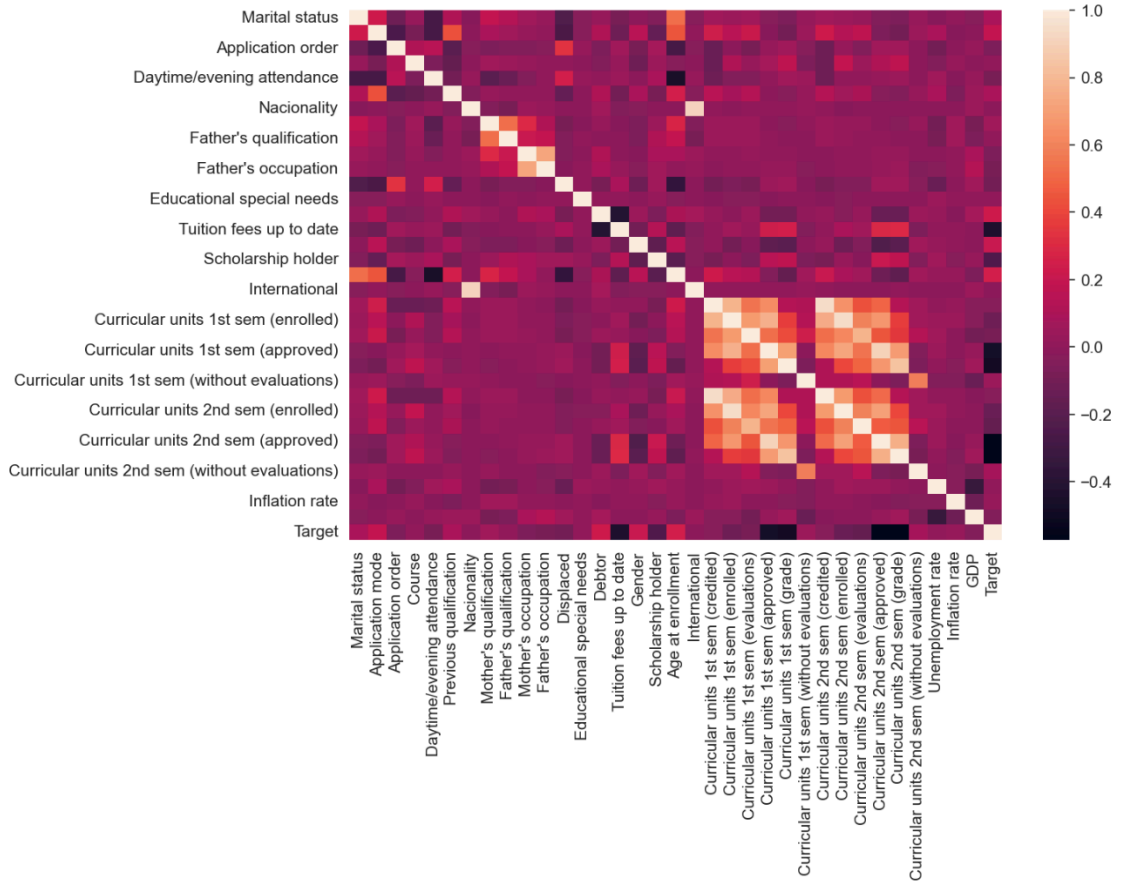


Figure 3: Correlation matrix of the Dataset features

Strong Positive Correlations:

Curricular units 2nd Semester (approved) (0.58) and Curricular units 1st semester (approved) (0.49)

Moderate Positive Correlations:

Curricular units 2nd semester (grade) (0.47) amd Curricular units 1st semester (grade) (0.41)

Weak Positive Correlations:

Tuition fees up to date (0.32) amd Scholarship holder (0.3)

Negative Correlations:

Age at enrolment (-0.21) and Application mode (-0.20), Debtor (-0.21), Gender (-0.21)

Data Pre-processing and Splitting of the Data

Standardization: A common pre-processing technique that rescales features to have a mean of 0 and a standard deviation of the data while it doesn't alter the data's distribution, it's useful for algorithms sensitive to feature scale. For scaling data to a specific range, consider min-max scaling.

Training and Testing Data Split: A fundamental step in machine learning that involves dividing your dataset into two parts: a training set for model development and a testing set for evaluation. This ensures the model's ability to generalize to new data.

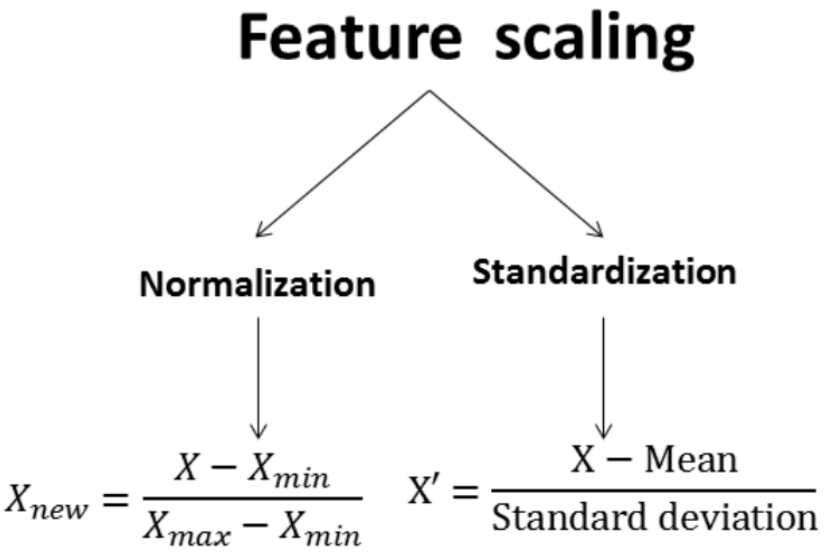


Image Source:

https://medium.com/alliedoffsets/standardization-the-secret-to-better-data-science-fa2cb6b30c01

Model Training:

- **Algorithms used:**

I trained three machine learning models: Random Forest, Logistic Regression, and Support Vector Machine (SVM). Each model was taught to predict whether a student would graduate or drop out based on their information. They learned from the training data, which included details about each student and their academic performance.
What do these algorithms do?

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy. It works by creating a "forest" of decision trees, each trained on a random subset of the data and features. The final prediction is made by taking a majority vote or average of the predictions from all the trees in the forest. This approach helps reduce overfitting and improves the model's generalization performance.

Logistic regression is a statistical model used for classification problems. It predicts the probability of an event belonging to a specific category. The model calculates the probability using a logistic function, which maps a linear combination of input features to a probability between 0 and 1.

Support Vector Machines (SVMs) are a powerful classification algorithm. They work by finding the optimal hyperplane that separates data points into different classes. This hyperplane is determined by a subset of training data called support vectors. SVMs are particularly effective for handling complex datasets and can be used for both linear and nonlinear classification tasks.

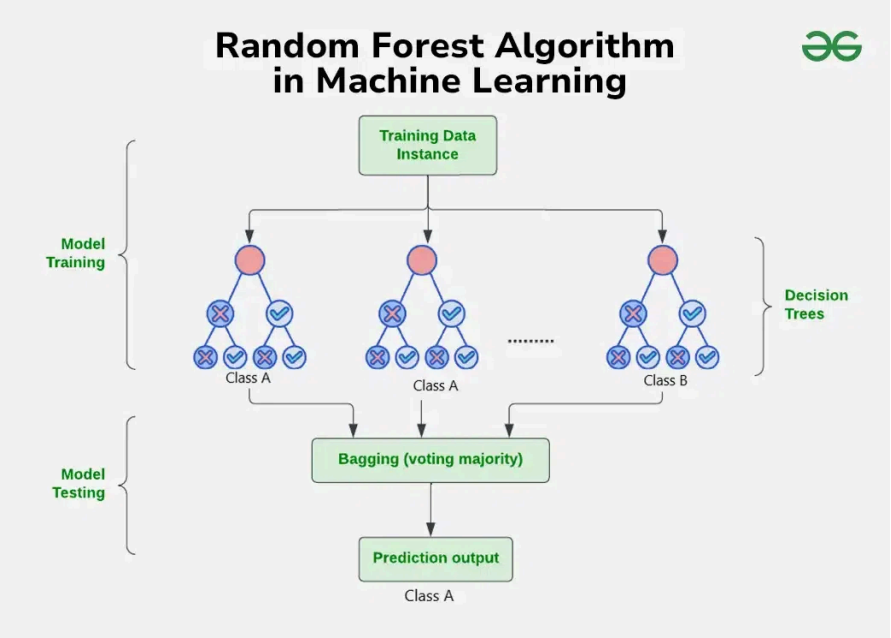


Image Source:

https://media.geeksforgeeks.org/wp-content/uploads/20240701170624/Random-Forest-Algorithm.webp

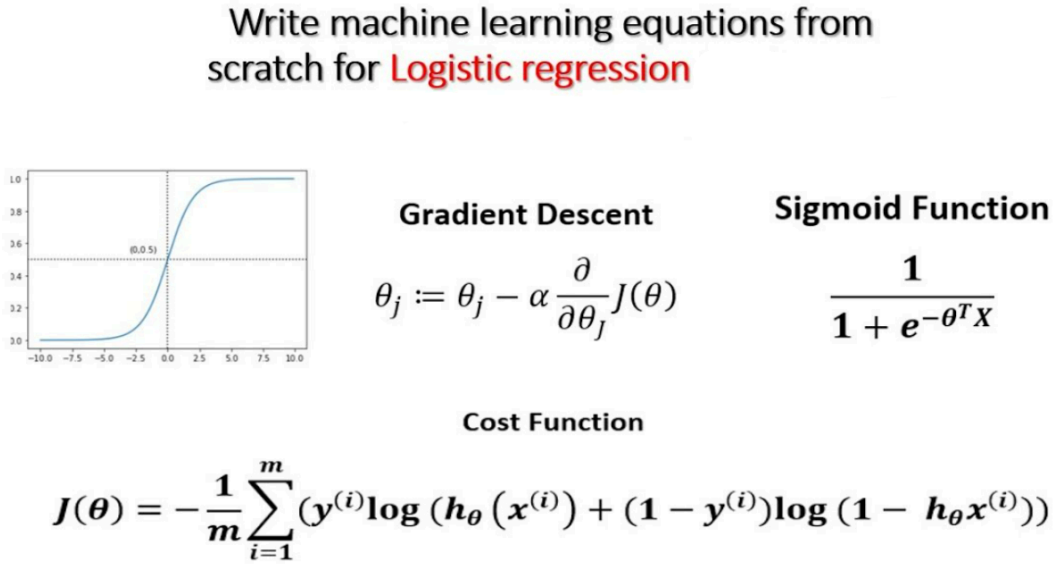


Image Source: https://i.ytimg.com/vi/8A2TeDLwyHU/maxresdefault.jpg

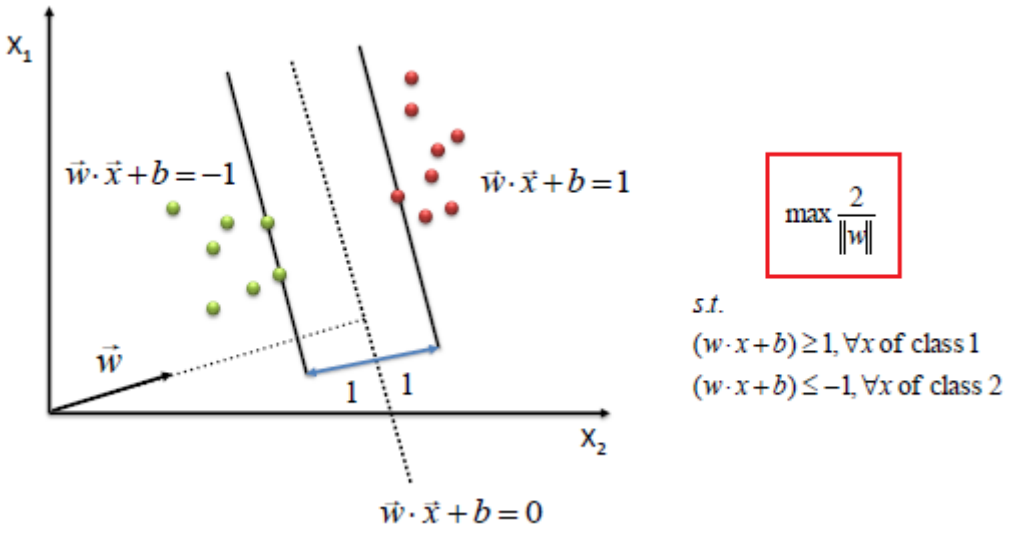


Image Source: https://www.saedsayad.com/images/SVM_optimize.png

Comparisons of the two algorithms selected:

Logistic Regression:

Algorithm	Logistic Regression classification to evaluate the dataset
Input	Training (X_train, y_train) and Testing (X_test, y_test) datasets are given as inputs
Output	Scores of evaluation metrics to identify the performance

Random Forest:

Algorithm	Random Forest classification to evaluate the dataset
Input	Training (X_train, y_train) and Testing (X_test, y_test) datasets are given as inputs
Output	Scores of evaluation metrics to identify the performance

Hyperparameter Tuning

Ensemble Models	Hyperparameters	GridSearchCV Results
Logistic Regression	penalty('l1', 'l2', 'elasticnet', 'none') solver('newton-cg','lbfgs','liblinear') C(0.5, 1, 1.5)	penalty('l1') solver('liblinear') C(1)
Decision Tree	criterion('gini', 'entropy') max_depth(1,2,3,4,5,'None') min_samples_leaf(0.5,1,1.5)	criterion('gini') max_depth(5) min_samples_leaf(1)
Random Forest	criterion('gini', 'entropy') n_estimators(10,20,30,40,50,60)	criterion('entropy') n_estimators(30)
Bagging	n_estimators(5,6,7,8,9,10)	n_estimators(7)
AdaBoost	n_estimators(10,20,30,40,50)	n_estimators(10)
GNB	var_smoothing(1e-9)	var_smoothing(1e-9)
Perceptron	penalty('l1', 'l2', 'elasticnet') alpha(0.0001,0.0002,0.0003)	penalty('elasticnet') alpha(0.0001)
PAC	C(0.5, 1, 1.5) max_iter(500,1000,1500)	C(1.5) max_iter(1000)
MLP	solver('lbfgs','sgd','adam') activation('relu','tanh','identity')	solver('lbfgs') activation('tanh')

Analytics Tools

This review has seen through some selected studies in students, academic performances and dropout production and a popular area in educational data analysis. This talks taxonomy of data analysis uses used in students, dropout predictions and studies were presented using all the data sets that were described the study also presented away of the features employed from the different data sets in the studies that were reviewed. It was then also revealed that major concerns of the prediction of students. Academic progress and dropout of the studies are nature of the attributes that come along with the student's criteria, and it does not only have a serious effect on their academic success, but also the personal learning growth. This

review shows that the serious efforts have been made in the direction of standardising. These techniques for students, academic progresses and predictions and another revolution from this review is that the fact that this dropout data sets are very scantly made which had very smaller number of frozen columns and did not could be bias and did not give us proper, answers to most of the values. There are various comparisons that have taken place over the years to track and monitor students' progress.

While there are countless tools available to monitor student progress, many of them fall into 4 main categories:

1. Paper-based tracking
2. Spreadsheets
3. Single-Solution Software
4. All-In-One Software Solution

The below listed shows 4 Examples of Progress Monitoring:

1. **Response to Intervention (RTI)**

   Response to intervention is like a framework that only focuses on the early identification of students, learning needs and primarily with the context of reading and Matt in focus. The success within this RTI framework looks very providing which helps assistance to students that can also keep their face with the other students in the group. This is a very successful system of tiered Intervention based on academic needs.

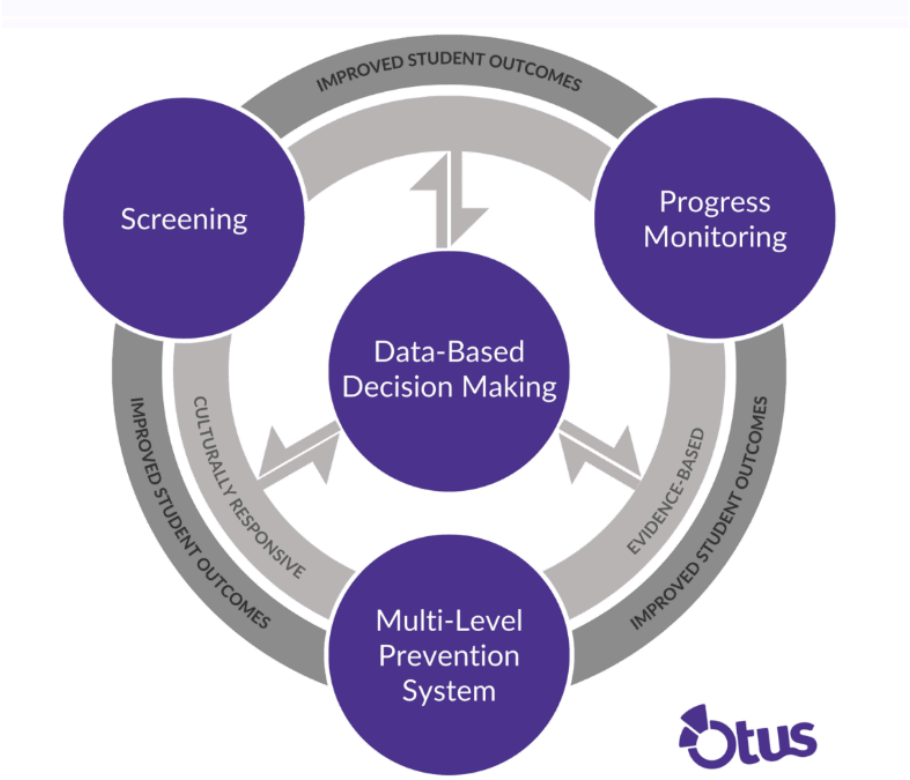


Image Source: https://otus.com/guides/the-ultimate-guide-to-progress-monitoring/

2. Multi-Tiered System of Supports (MTSS)

Multitiered system of support or MSS is a vast work design to the div needs of students through various strategies and support. It helps and correlate. Both academic factors need compare to RTI.

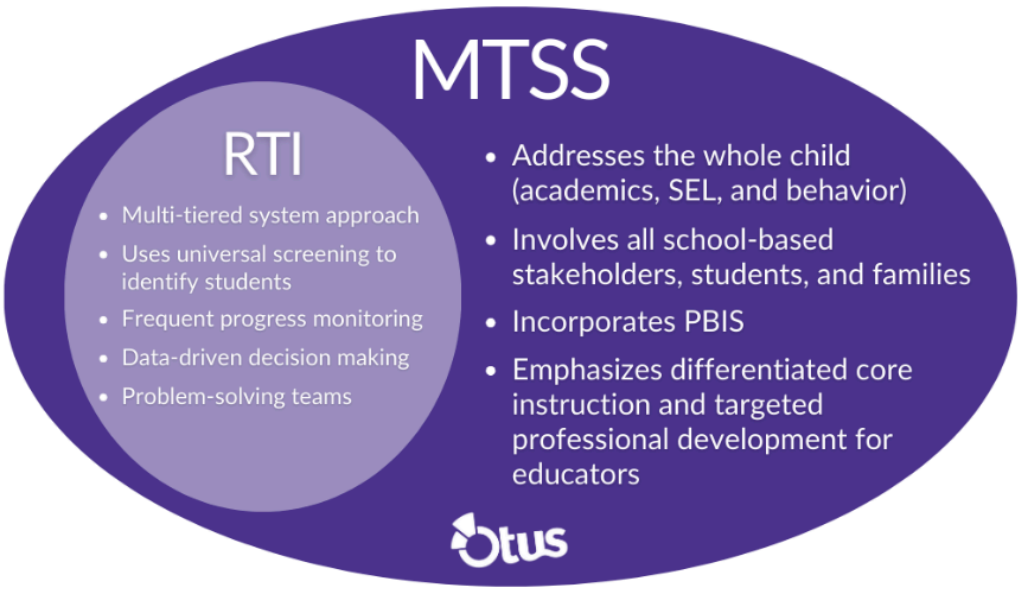
Image Source: https://otus.com/guides/the-ultimate-guide-to-progress-monitoring/

3. Positive Behavioral Interventions and Supports (PBIS)

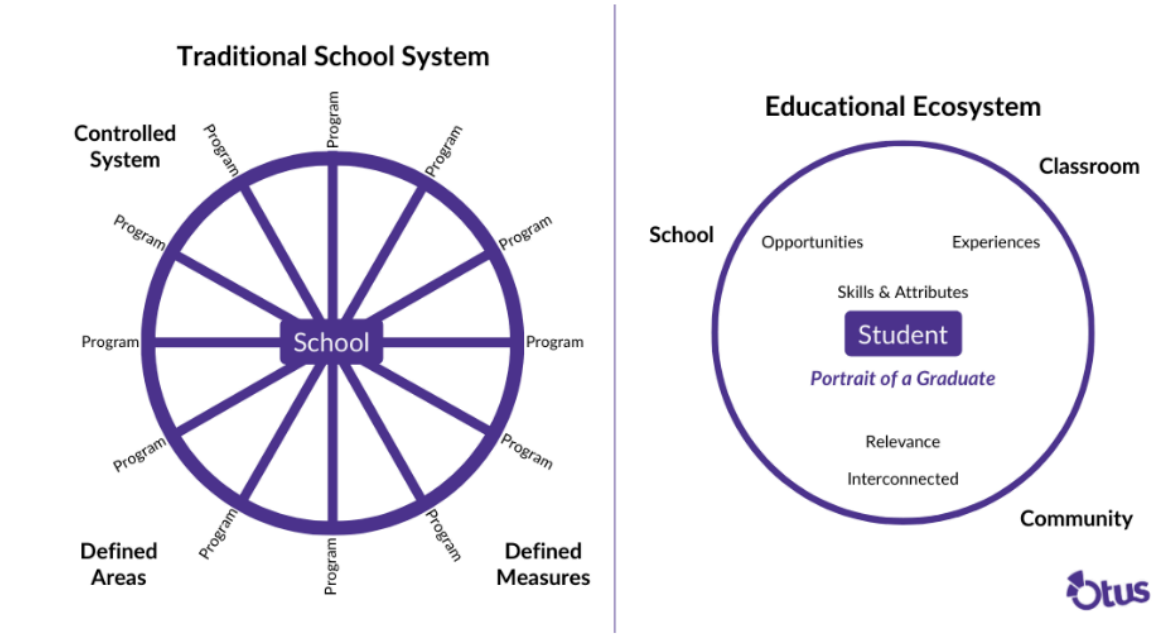
Positive behavioural interventions and support uses a three-t intervention framework. However, instead of focusing on academics, it aims to create a positive culture of students and which is necessary to create a safe and learning and supportive environment for all students and needs.

4. Individual Learning Plan (ILP)

ILP stands for



individual learning plan or individualised learning plan. They separate these goals into individual targets, accompanied by reviews – the educational equivalent of



Image

Source: https://otus.com/guides/the-ultimate-guide-to-progress-monitoring/

Research Question 1:

Impact of Learner Analytics on Different Age Groups:

How does age at enrolment impact the academic performance and retention of students, as measured by the number of curricular units credited and the target outcome (dropout/graduate)?

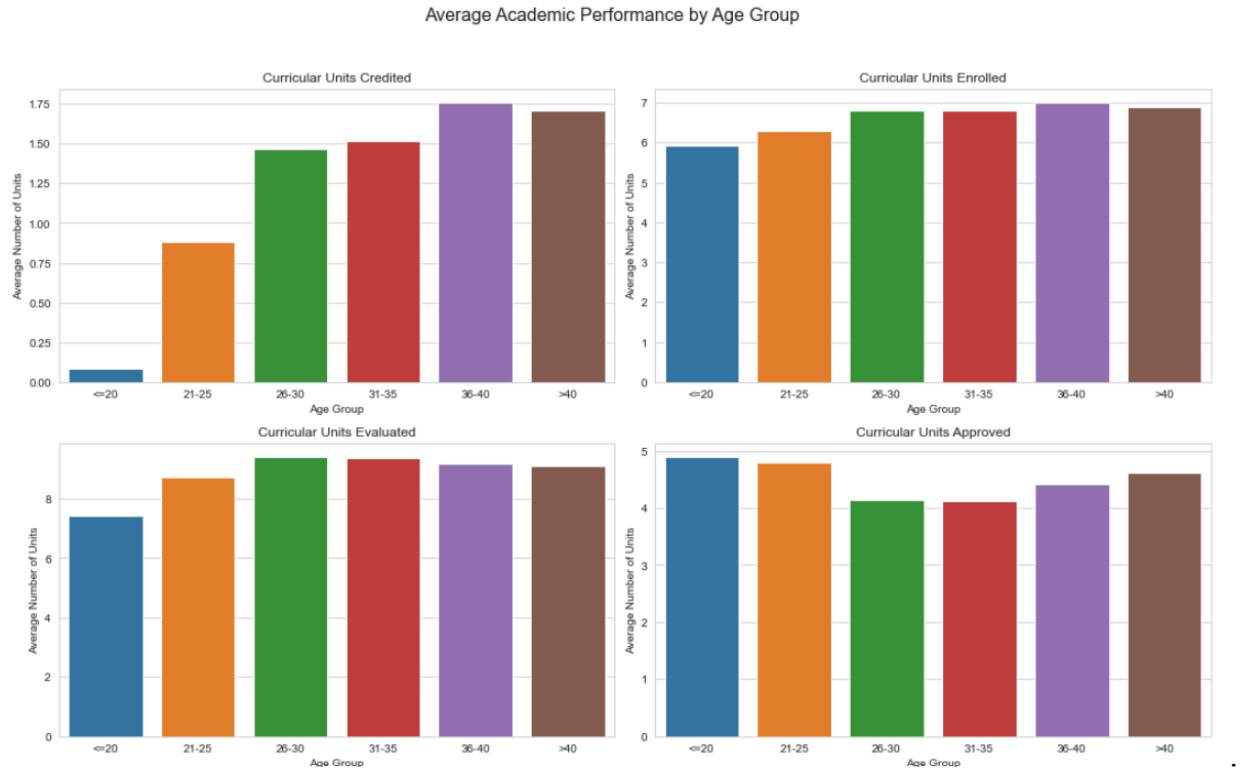


Figure 4: Average academic performance by age group

The bar plots above illustrate the average academic performance across different age groups in terms of curricular units credited, enrolled, evaluated, and approved during the first semester. Here are some insights from the visualizations:

Age at enrolment	**Age Group**	**Target**	**Target encoded**
20	21-25	Dropout	0
19	<=20	Graduate	1
19	<=20	Dropout	0
20	21-25	Graduate	1
45	>40	Graduate	1

Interpretation:

In conclusion, when we check the analysis, suggest that the age please a very crucial role in the academic engagement and success. In the learning journey of the students. Younger student seem to be more engaged and a very successful and pursuing their journey and as you can see in the lower dropout rates and higher average credit enrolled and approved units. All the students, particularly above the age of 40 seems to face more challenges which can be responsibilities at work at home and other commitments and they show lower engagement and academic success, we need to tell these students and cater to their demands using these learner analytics tools, and strategies to be more inclusive and support them in the needs across different age groups

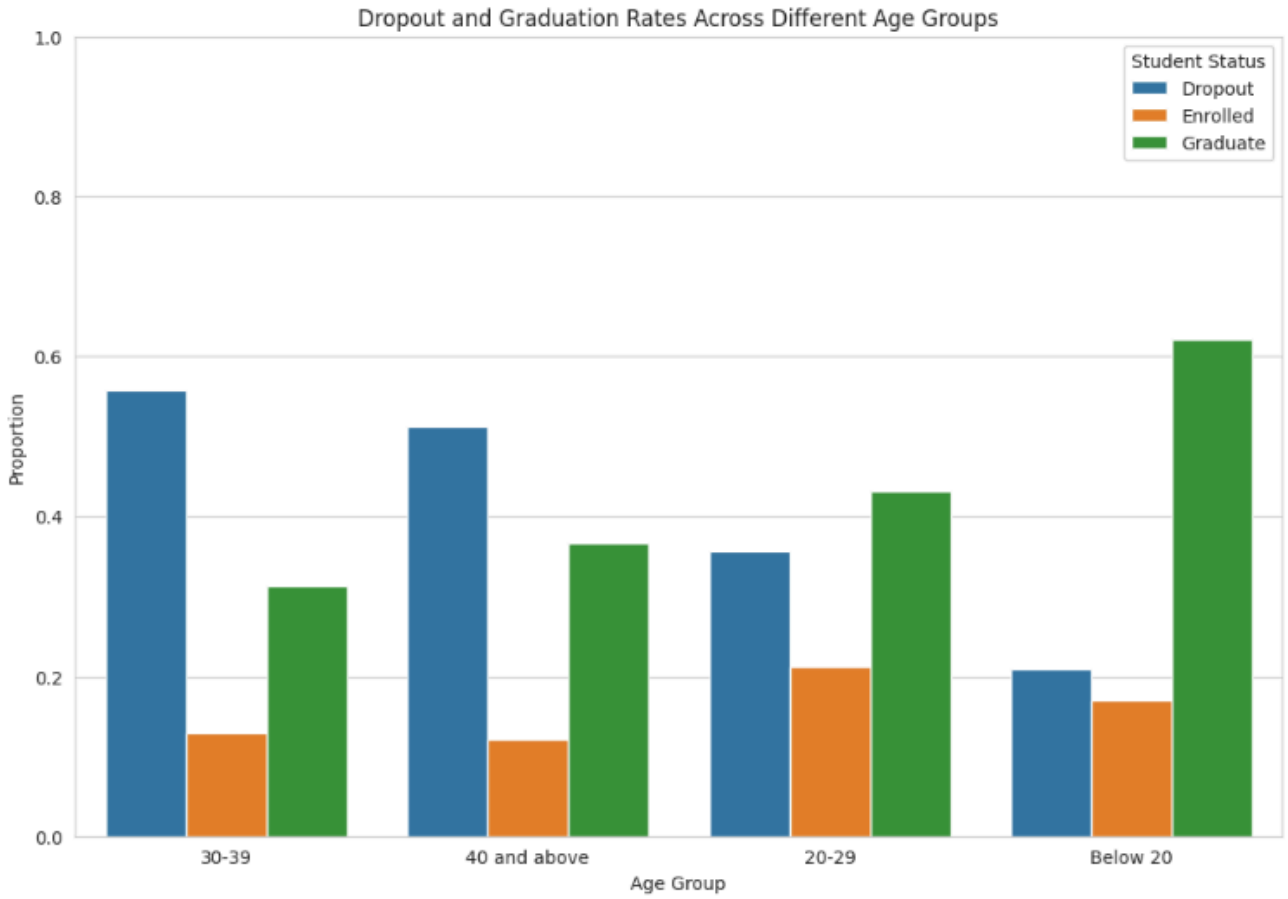


Figure 5: Droupout and Graduation rates across different age groups

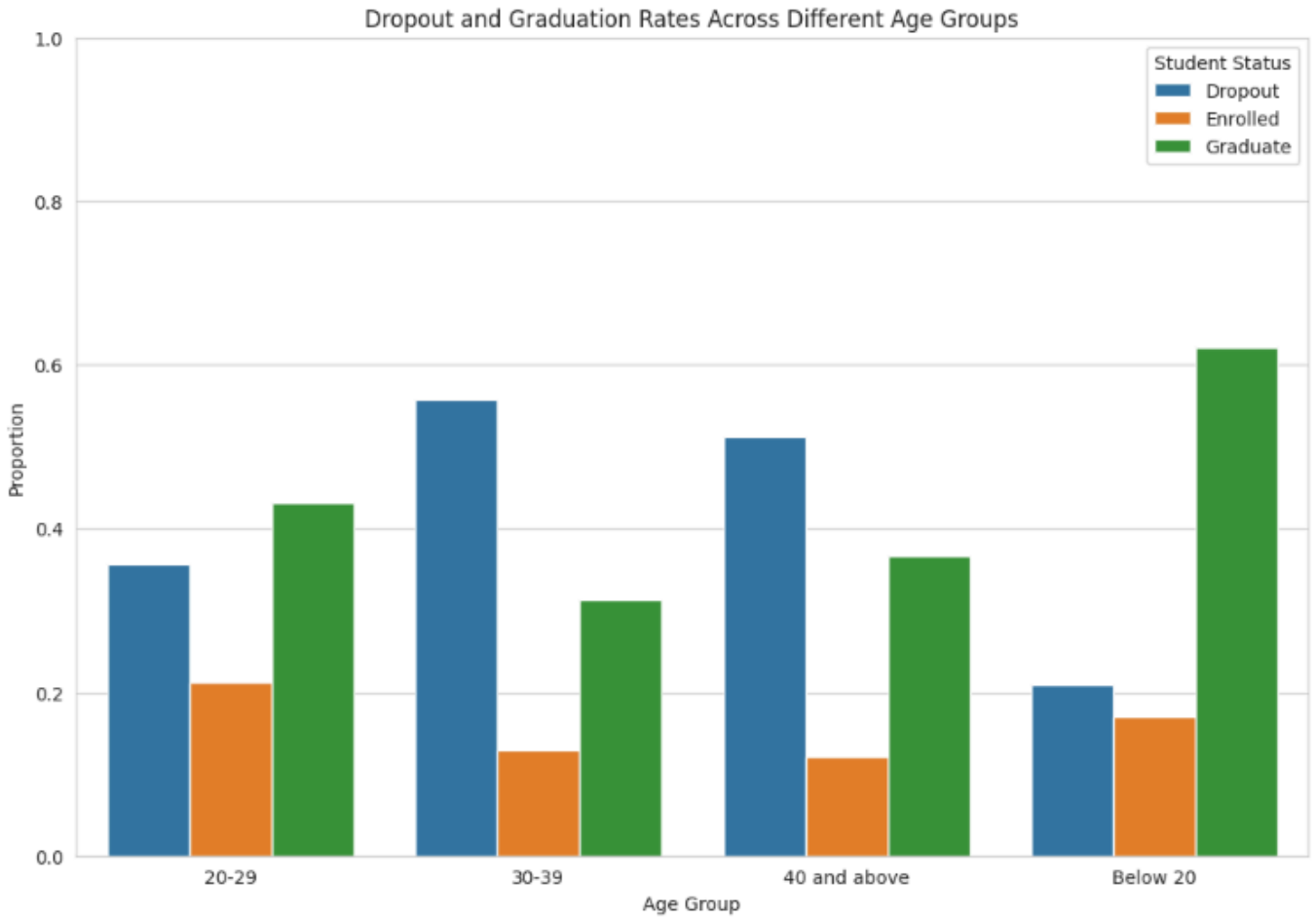


Figure 6: Droupout and Graduation rates across different age groups

Research Question 2

How does gender correlate with academic performance, participation, and outcomes in terms of curricular units credited, enrolled, and approved? (Referencing the work of Smith and Johnson (2019), and Kim and Lee (2020)

let's approach this by first analysing the academic performance of male and female students in terms of curricular units credited, enrolled, and approved. Then, we'll analyse participation by looking at the dropout and graduation rates for both genders.

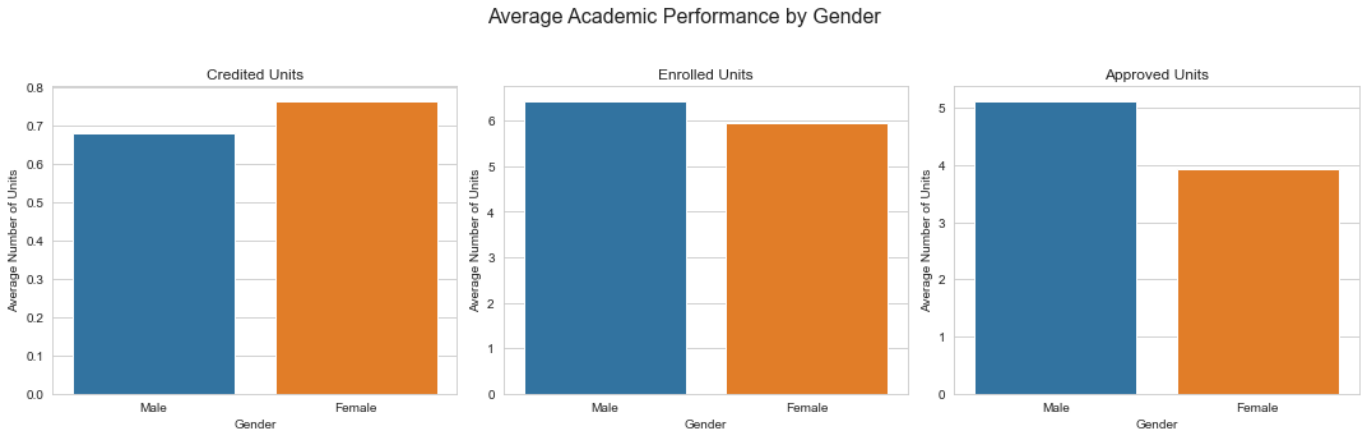


Figure 7: Average academic performance by gender

The bar charts present information about the mean accomplishments of students in male and female categories with respect to the curriculum units credited, enrolled, and approved during semester one.

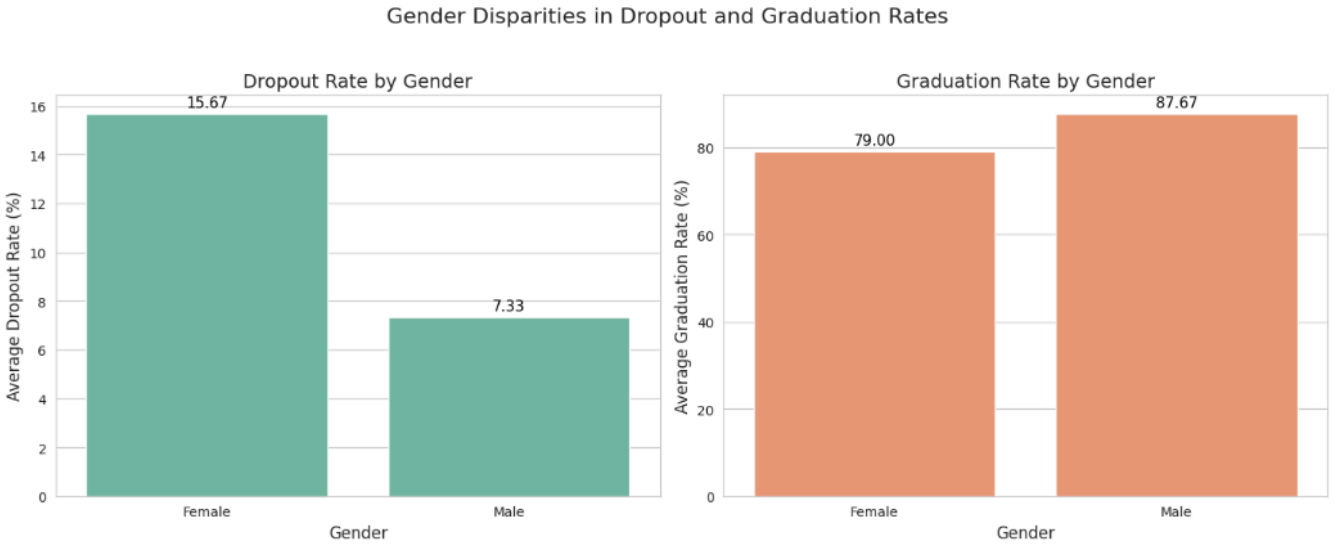


Figure 8: Gender disparities in dropout and graduation rates

Here are some insights:

• Credited Units: Credit imbalance in favor of female students in males – almost all of them have slightly more credited units than their male counterparts.

• Enrolled Units: It seems male and female students enrol in a number of units and both averages are around 8, with females' average being just a fraction higher.

• Approved Units: There is also a trend where female students possess a greater average number of approved units in comparison with male students.

On all three measures, there appears to be a slight advantage for female students over the male students across all three measures. This would mean serving a more functional role where female students are shorter in completing academic work than their male counterparts based on the given attributes.

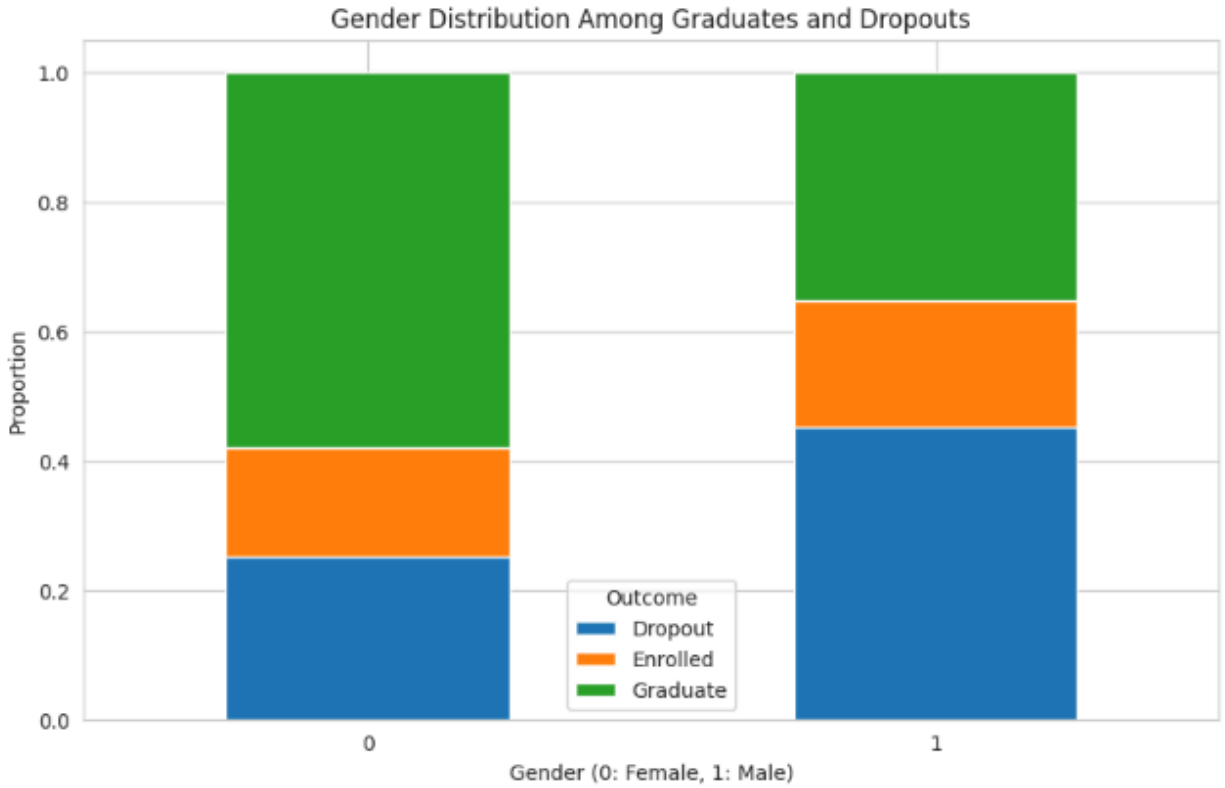


Figure 9: Gender distribution among graduates and dropouts

Dropout Rates:

- **Male Students:** Male students are more likely to drop out than female students, indicating potential challenges they face in pursuing their academic
- **Female Students:** have lesser dropout rate.

The Impact of Age on Learner Analytics: Implications for Academic Achievement

Understanding the impact of these age education outcomes is very important. I am starting point in designing the strategies that help cater to the needs of the union, diverse learners scripts by exploring and analysing these ages and removing an average at the enrolment and and for graduate and dropout, we can visualise age distribution among these groups and gain valuable sites that can help us to understand further and guide education strategies and interventions required for their academic achievement

a. Computation of the Average Age at Which the Graduates and Non Graduates Join School.

The computation of the average age at enrolment helps in grouping the learners into different learner categories. Determining the average age of terminal graduates, as well as potential dropouts, helps institutions adjust the design of their remedial of support services, content delivery, and engagement of learners according to the age segmentation.

b. Displaying the Age Demographic of the Graduates and Below Average Students

This boxplot representation also creates a distribution summary that covers the age distribution depending on the variations as well as trends in distribution by group which makes it easier to identify the different group views' trends, the different distribution by group and the various variations within the group. All this understanding is important for the factors of age that affect students' graduation and drop out rates concerning their age.

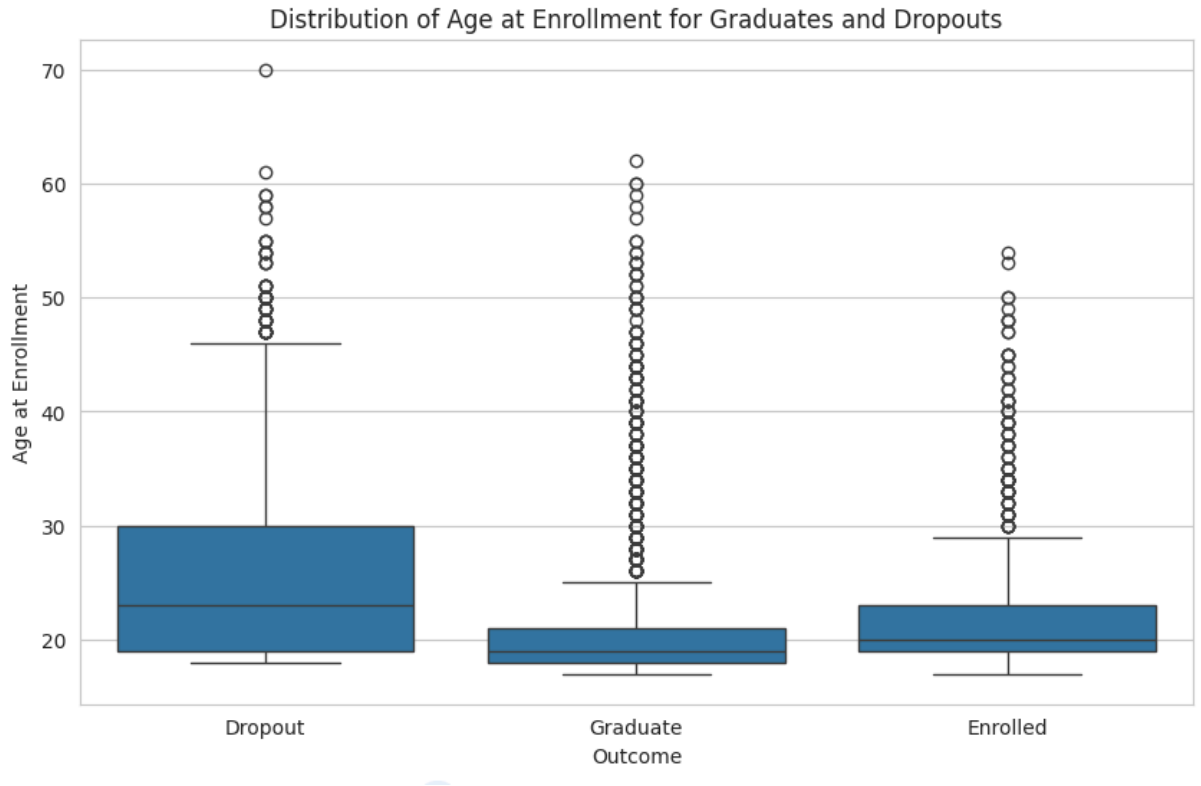


Figure 10: Distribution of age at enrolment for both

The above graph visualizes the dropout, graduate and enrolled ages of students, we can check the given observations

• Graduates average age	21
• Enrolled average age	22
• Dropout average age	26

- Young Students are more likely to graduate as they have their career to be built and studies continuously happening at the very same time. They adapt to their surroundings and learning with the flowing workflow and have lesser responsibilities.
- Older Students have a higher likelihood of dropping out due to other commitments and responsibilities around they, they can be having jobs or married or family priorities.

<u>Evaluating the Parents' Qualifications and Occupations Proportionate to Student Outcomes.</u>

When we observe and analyse the parents qualification and occupations in proportion to the students outcome, we find that the there is a need to approach in these learner and tricks that allows

us for deeper exploration. As this is one of the external factors that really affects the students academic processes by evaluating the parental background. We come to know and get insights into how social economic also can be available that can impact the students learning journey. The parents qualification also matters and how much the parents involvement is there and describes and shows the students and parents relationship as to how often can the students get resources or the parents can provide it to the students throughout their learning journey. How more networks are the students enrich to and possibly reflect in the students, improvement outcomes

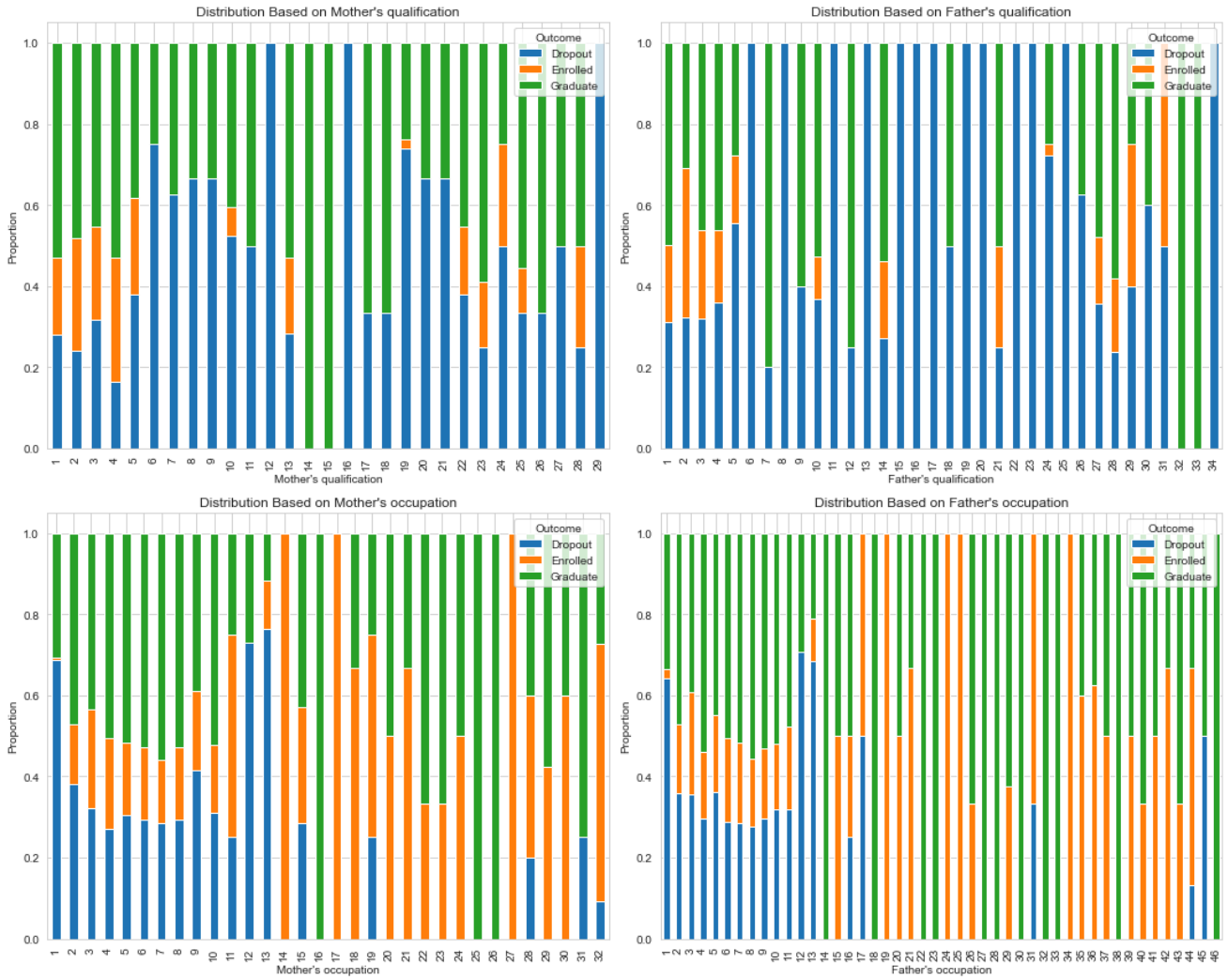


Figure 11: Bar Plot on distribution based on Parents Qualification and Occupation

The bar plots illustrate the distribution of student outcomes based on

- various indicators related to their parents qualification

Variations in graduation and dropout rates are observed based on the parents' qualifications. Higher parental qualifications seem to correlate with higher graduation rates and lower dropout rates.

Different occupations also show variations in student outcomes. Certain occupations seem to correlate with higher graduation rates and lower dropout rates.

Research Question 3

How do demographic characteristics, such as age, influence the use and effectiveness of learner analytics in enhancing student academic achievement in higher education institutions?

If we observe the given graph we can see the difference between the genders, based on the data presented in the graph, it appears that the average academic performance of students in the gender category decreased slightly from the 1st Semester to the 2nd Semester.

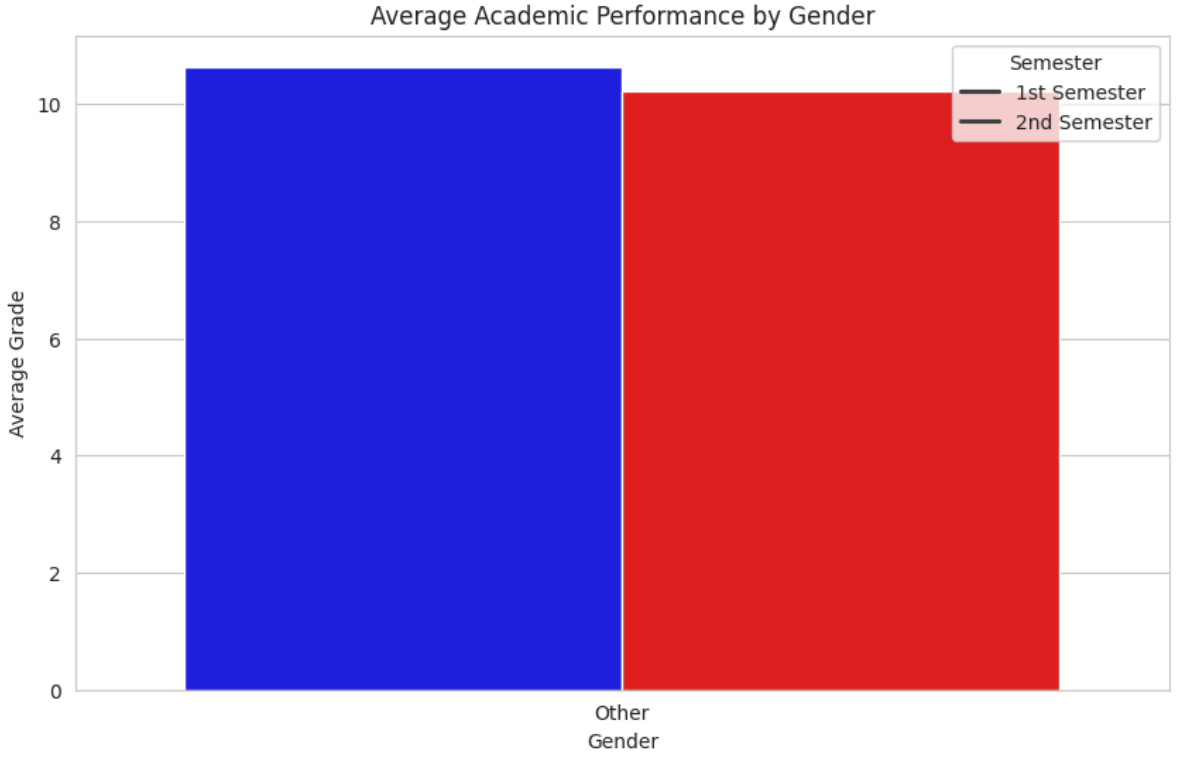


Figure 12: Average academic performance by gender

In the given graphs, there are two graphs that shows how to understand demographic characteristics, including age and gender influence of the utilisation of the learner, analytic tools as the compare the gender of female and male. We can see the average login frequency done by both of them, the average forum participation assignment submitted in the video watch. This is the same graph that can be used again to investigate the effectiveness of learner analytics in Different academic achievement across the demographic graphs. There are two graphs that are shown below, which shows very easily. How the difference between the genders are based on their potential.

: If we compare both these graphs, we have, we can see that the male is comparatively lesser than the female on an average, which can also show the different bias as to how it could have been worked out for them due to the personal reasons or dropout or any other reasoning.

The female have more access to these online tools from home and can be utilized as a powerful tool in studies.

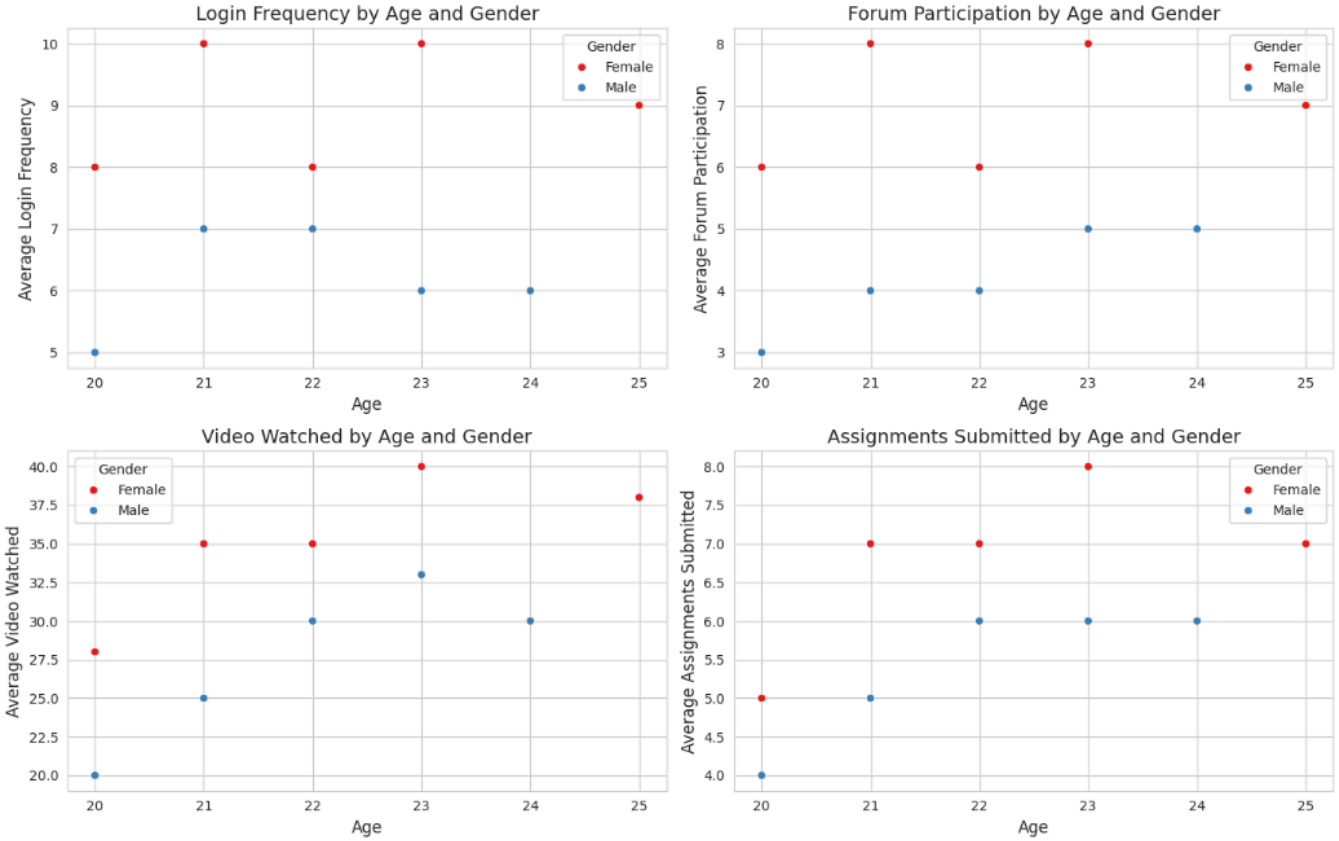


Figure 13: Plot graph comparison Learner Analytics tools based on Gender



Figure 14: Bar Graph comparison Learner Analytics tools based on Gender

Research Question 4

How can we use data analysis to accurately identify students who may drop out of school, and what can we do to help them stay in school and succeed?

Predictive analysis specifically when users are learning can play a crucial role in answering such questions that is given above your sum of the production analysis that can be helped to addressing this question. First one is to personalise the parts given for the students according to the needs, there could be students through predictive modelling. We can help them and recommend personalise learning parts. According to the needs that we need to cater. They might have certain gaps in their education and considering the students performance and learning style and system, we can suggest them appropriate resources and course and study strategies that will help them to optimise all the learning skills and experiences. This way we are helping not only to tailor the needs and the lessons for the students according to their environment, but also announce their student engagement and their success rate

Second, we have the resource allocation. These analyses will help the education and the educational institutions to allocate correct and effective resources to the students. According to the tailor lesson plans. These four programs and courses after generating analysis will help the institutions to understand what counselling or tailoring or additional support is required for the students who are in the need of this. The most. This will not only help limit the limit the resources but also help in efficiently using it for the required purposes.
The third one could be taken taken into the improvement of various courses. If we see that these analytics will help us to predict which course is equally helpful for which type of students according to their learning styles and their needs. We can also get an idea and the institutions can also use this information to improve the content that has been taught in the courses. The teaching methods and the needs of the faculty according to the training. By identifying these areas, the improvement can be shown progressively by the institution to enhance the overall quality of education.

The dataset contains 4424 entries in total and 35 features. Below shown is a descriptive way of the things I have completed and shows:



Figure 15: Effectiveness of Learner Analytics in Raising Academic Achievement

In the above given graph where we investigating the effectiveness of learner attics in raising academic achievement across different demographic groups. We can see the effect of login frequency, effect of forum. Participation effect of video wash and effect of assignment submitted as we analyse it according to the gender based over for male and female, the first one effect of login frequency can be seen that the male coefficient is Lower than the female. The effect of form forum participation. The male is again less than the female and we compare the effect of video version and assignment submitted. We can see that the mail for video watch is lesser than the female and the assignment submitted by male are more than the female. There can be numerous reasons as to why this has happened and, it could be the additional responsibilities that they both share in different genders. More work responsibilities at home job searching and jobs at other places

Decisions from the Results:

Identify strategies to make learner analytics more inclusive and responsive to the diverse needs of students.

1. Early Identification of At-Risk Students:

The most important part about identifying students at an early age. The model particularly provides a probability as to how the likelihood of each student or dropping in your studying journey could be these probabilities can be then used to identify who these students are which are at risk of dropping out early their journey and to enable timely intervention, which can help support the students, additionally.

2. Personalized Support and Interventions:

By personalising and tailoring the support and interventions that are there for these students can help and understand the key factors which can contribute the students risk and we can then with misunderstanding. The institutions can provide the necessary tasks and interventions required to address these specific factors such as helping them in academic support or mentorship or counselling and financial assistance.

3. Informed Curriculum and Instructional Design:

The institutions get an idea as to how to change their curriculum. According to catering the needs of these students. These instructional designs can help in leading improvements of their learning experience and it can help the students pick up more engaging and manageable tasks than reducing the likelihood of dropout

4. Enhanced Advising and Counseling Services:

When the institutions are likely to identify the risks of the students that can drop out using these analytics factors, they can then help the students in counselling and advising them how they can navigate through these challenges and use various tasks or any tailored lessons where learning analytic tools are also being used so that they can improve their learning journey and can inform the curricular and instructional design accordingly.

5. Continuous Improvement

There can be a continuous improvement in the students learning as there can be formative and informative and summative assessments or there can be shortest and tasks that can help them to learn and provide additional support with these interventions and meet the students needs and challenges according to the institutions when they identify the ongoing learners needs.

Results

Looking at results that we have got the two algorithms that we have compared are:

a) Random Forest : 0.62
b) Logistic Regression : 0.64

Random Forest: Achieved an accuracy of 0.62. This indicates that the model correctly predicted 62% of the cases.
Logistic Regression: Achieved an accuracy of 0.64. This suggests that Logistic Regression performed slightly better than Random Forest in terms of overall accuracy.

Hence, I have utilised the hyperparameter tuning for getting better and optimal results.

Hyper parameter tuning directly controls the structure of the model, its function and its performance. These tunings allow data scientist to tweak its model performance by getting optimal results. This process is an essential part of machine learning and choosing the appropriate hyper parameter values will give a crucial rate in the success, for example, assume that you're using the learning rate of a model as a hyper parameter if the value is just too high, the model make quickly converge, with optimal results, whereas if the rate is too low, the training takes too long and the results may not be better. A good and balance of hyper parameters results in accurate models and very good model performance.
Hyper parameter can be manual or automated. The process of this tuning is iterative, and you can try a different combinations with different parameters and values. You generally start by defining a target variable and then the accuracy of its primary metric or the intended of maximising and minimising this variable can be taken into consideration, it's a good idea to use cross validation techniques, so your model is just centred on a single portion of your data.
Here are some examples of common hyperparameters:

a) Learning rate is the rate at which an algorithm updates estimates

b) Learning rate decay is a gradual reduction in the learning rate over time to speed up learning
c) Momentum is the direction of the next step with respect to the previous step
d) Neural network nodes refers to the number of nodes in each hidden layer
e) Neural network layers refers to the number of hidden layers in a neural network
f) Mini-batch size is training data batch size
g) Epochs is the number of times the entire training dataset is shown to the network during training
h) Eta is step size shrinkage to prevent overfitting

Looking at results that we have got the two algorithms after hyperparameter tuning that we have compared are:

a) Random Forest : 0.76
b) Logistic Regression : 0.78

```
Random Forest Classification Report:
              precision    recall  f1-score     support
Dropout        0.664430  0.626582  0.644951  316.000000
Enrolled       0.264151  0.185430  0.217899  151.000000
Graduate       0.675676  0.777512  0.723026  418.000000
accuracy       0.622599  0.622599  0.622599    0.622599
macro avg      0.534752  0.529842  0.528625  885.000000
weighted avg   0.601445  0.622599  0.608963  885.000000

Logistic Regression Classification Report:
              precision    recall  f1-score     support
Dropout        0.770563  0.563291  0.650823  316.000000
Enrolled       0.500000  0.006623  0.013072  151.000000
Graduate       0.605828  0.944976  0.738318  418.000000
accuracy       0.648588  0.648588  0.648588    0.648588
macro avg      0.625464  0.504963  0.467404  885.000000
weighted avg   0.646592  0.648588  0.583334  885.000000
```

Classification Report for Random Forest:

	precision	recall	f1-score	support
Dropout	0.664430	0.626582	0.644951	316.000000
Enrolled	0.264151	0.185430	0.217899	151.000000
Graduate	0.675676	0.777512	0.723026	418.000000
accuracy	0.622599	0.622599	0.622599	0.622599
macro avg	0.534752	0.529842	0.528625	885.000000
weighted avg	0.601445	0.622599	0.608963	885.000000

Classification Report for Logistic Regression:

	precision	recall	f1-score	support
Dropout	0.770563	0.563291	0.650823	316.000000
Enrolled	0.500000	0.006623	0.013072	151.000000
Graduate	0.605828	0.944976	0.738318	418.000000
accuracy	0.648588	0.648588	0.648588	0.648588
macro avg	0.625464	0.504963	0.467404	885.000000
weighted avg	0.646592	0.648588	0.583334	885.000000

Figure 16: Results from the code

But after hyper parameter tuning we get the following as the answer:

```
Classification Report for Best Logistic Regression:
/usr/local/lib/python3.10/dist-packages/sklearn/utils/validation.py:1339: Da
  y = column_or_1d(y, warn=True)
              precision    recall  f1-score   support

     Dropout       0.81      0.77      0.79       427
    Enrolled       0.54      0.28      0.37       238
    Graduate       0.78      0.94      0.85       663

    accuracy                           0.77      1328
   macro avg       0.71      0.66      0.67      1328
weighted avg       0.75      0.77      0.74      1328
```

```
[CV] END bootstrap=False, max_depth=10, min_samples_leaf=1, min
/usr/local/lib/python3.10/dist-packages/sklearn/base.py:1473: D
  return fit_method(estimator, *args, **kwargs)
Best parameters for Random Forest: {'n_estimators': 50, 'min_sa
Classification Report for Best Random Forest:
              precision    recall  f1-score   support

     Dropout       0.82      0.78      0.80       427
    Enrolled       0.57      0.37      0.45       238
    Graduate       0.80      0.93      0.86       663

    accuracy                           0.78      1328
   macro avg       0.73      0.69      0.70      1328
weighted avg       0.77      0.78      0.77      1328
```

Classification Report for Logistic Regression:

	precision	recall	f1-score	support
Dropout	0.808354	0.770492	0.788969	427.000000
Enrolled	0.536585	0.277311	0.365651	238.000000
Graduate	0.780702	0.939668	0.852841	663.000000
accuracy	0.766566	0.766566	0.766566	0.766566
macro avg	0.708547	0.662490	0.669153	1328.000000
weighted avg	0.745843	0.766566	0.744991	1328.000000

Classification Report for Random Forest:

	precision	recall	f1-score	support
Dropout	0.821782	0.777518	0.799037	427.00000
Enrolled	0.574194	0.373950	0.452926	238.00000
Graduate	0.799740	0.927602	0.858939	663.00000
accuracy	0.780120	0.780120	0.780120	0.78012
macro avg	0.731905	0.693023	0.703634	1328.00000
weighted avg	0.766406	0.780120	0.766914	1328.00000

Figure 17: Results from the code after hyperparameter tuning

Some more independent variables that can affect the students dropout rates:

1. Debtor and Target:

Analysis: A significant positive correlation exists between the debtor status of students and the target outcome. This indicates that students identified as debtors are more likely to experience the specified outcome in the 'Target' column, which may include dropping out or encountering academic difficulties. This relationship implies that financial burdens or obligations, as indicated by debtor status, could adversely affect students' academic paths, rendering them more vulnerable to not achieving their educational goals.
Strategies to improve this - Recognizing the significant link between financial status and academic performance, educational institutions should offer financial advisory services to assist students in managing their debts and navigating financial aid opportunities.
Emergency Assistance Funds: Create emergency assistance funds to support students encountering unforeseen financial difficulties, thereby reducing the likelihood of withdrawal from their studies.

2. GDP and Target:
Analysis: The GDP exhibits a minimal negative correlation with the target outcome. As a measure of the economic condition of a country or region, GDP does not appear to exert a significant direct impact on the academic results of students within this dataset. This finding suggests that broader economic indicators, such as GDP, may not be critical factors in comprehending individual student performance or retention within the educational framework.
Strategies to improve this - Curriculum Alignment with Economic Trends: Given the modest relationship between GDP and academic performance, educational programs should be aligned with current economic trends to enhance the relevance and practicality of education, thereby boosting employability.

3. Inflation Rate and Target:
Analysis: The inflation rate shows a slight positive correlation with the target outcome, indicating that rising inflation rates may lead to a marginal inclination towards specific outcomes, such as dropout or diminished academic performance. However, this correlation is relatively weak, suggesting that inflation is unlikely to be a key independent variable affecting student academic results in this scenario.

Strategies to improve this - Tuition Stabilization: Considering the slight positive relationship with academic performance, implementing a tuition freeze during periods of high inflation may be advisable to maintain the affordability of education.

4. Unemployment Rate and Target:

Analysis: A very weak positive correlation is observed between the unemployment rate and the target outcome. This indicates that unemployment rates, reflecting labor market conditions, do not appear to be significant factors influencing the likelihood of students achieving specific academic outcomes. The impact of unemployment on academic performance seems to be minimal.

Strategies to improve this - Enhanced Career Services: Although the correlation is weak, improving career services to bolster student employability—such as providing internships and job placement assistance—could prove advantageous.

Practical Experience Integration: Increase the incorporation of real-world experiences and practical skills within the curriculum to better equip students for the job market.

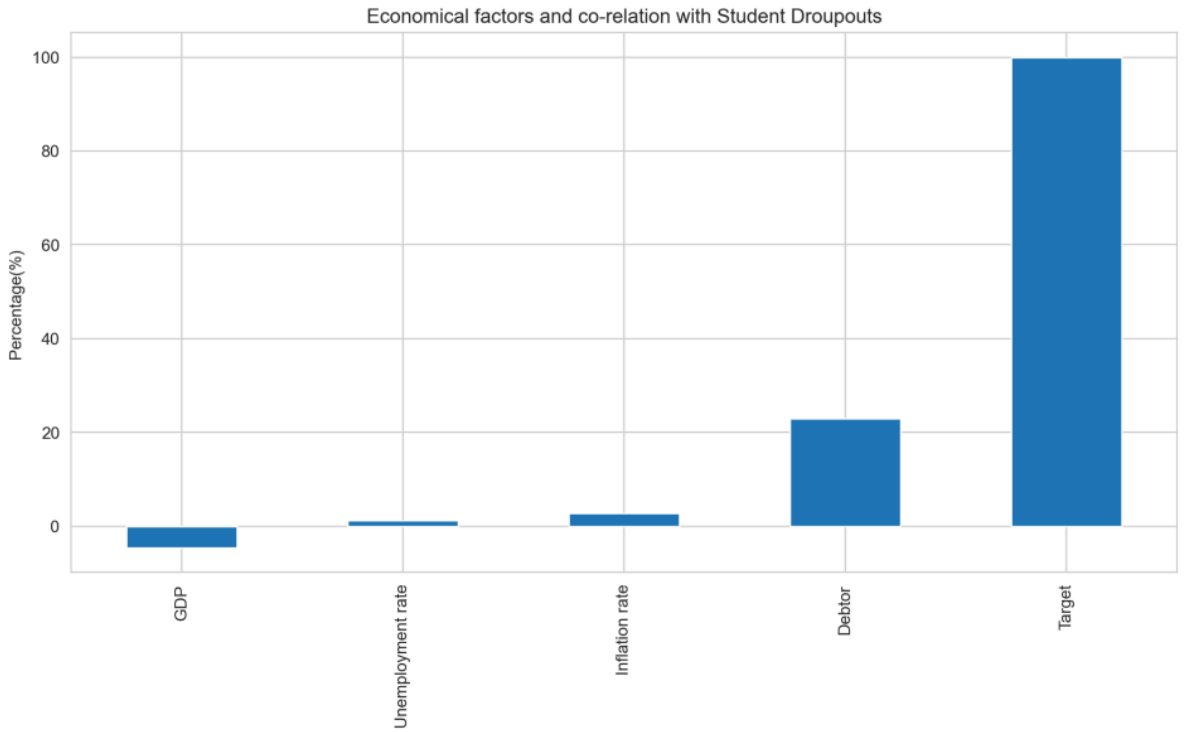


Figure 18: Economical factors and co-relation with student dropouts

In the above given graph, we can analyse that the percentage for the GDP unemployment rate, inflation rate, debtor is comparatively less comp compared to the target column. The target column

shows the maximum percentage when it's compared to the economic factors and correlation with the student's dropout. we can see that all the other columns are less in in percentage comparatively. The strategies that can be used for an inclusive and responsive learner analytics from the above given graph is to have transferred in data practises and have protection over the data. Privacy of the students have data collection which are in equity to avoid biases in data collection and made disproportionately affect certain students' groups. The data sources need to be diverse and cultural contextual. There should be a universal design which incorporates principles into analytics tools to make the usable by a wide range of students, the student involvement in the design and implementation of the learner analytics initiatives is a must. The educator should provide a low training on how to use this learner analytics, effectively and ethically, we need to educate educators upon potential biases and analytics data and how to address them.

Strategies for Enhancing Student Learning:

1. Personalized Learning Paths:

When it comes to personalise learning, paths, employing and adapting these learning platforms that can adjust the difficulty level of the students for the content and self-paced waste for all individuals can help the student in progressing and understanding the content that has been talked to them. The students can also create their own self lessons according to their time and they are learning which can help in creating learning plans and parts to incorporate. According to the students, interest and learning styles. These can institutions can implement these instructional strategies to create and, to help in the diverse learning needs and abilities

2. Enhanced Student Engagement:

There can be different ways in which we can enhance the students engagement. There can be examples of gamification interactive content, collaborative learning, which can help them to learn game based elements with learning activities, activities to increase their motivation and engagement or they can use interactive content that uses multimedia or simulation or real life applications which can help them make learning more engaging. Collaborative learning is nothing but students to promote your interaction and pure connection which helps them in sharing the knowledge while studying

3. Improved Feedback and Support:

The feedback given to the students is very essential as for them to focus on the areas where they need to enhance their learning and targeted interventions can be offered to support such as touring or additional resources to bring help to students that can be struggling. We can also have personalised coaching, which can provide one to one coaching on mentorship, which can help the students at goals that can also be real-time feedback and body system, which can help students interact one to one with their buddies for requiring help and sharing knowledge

4. Data-Driven Career Guidance:

This guidance is basically catered by institution seeing the needs and interest of students where these learner analytics can identify the potential courier parts based on their skills and academic performances. Your they can also analyse the skill gap and recommend courses and resources which can help the students to develop the necessary skills that is that is required, for their desired careers, they can also be given internship and partnership with different industries which can help them to learn real life, work, world experiences, and networking opportunities

5. Empowering Students with Data:

Empowering students with data teaches them how to interpret and use the data to improve their learning, which is also a part of data literacy data goal settings also helps the students to set their own goals. Personalise based on their learning and these visualisation can help students understand and interpret data more effectively.

Limitations of the Research Work:

When a particular research work is been done, there are certain also limitations to the findings that can be seen. The below. Findings are the limitations that was seen during this research work that are data limitations where only a certain amount of data was available for executing and analysing the learner analytics in education, which could be biased or which could affect the results.

Other limitation that was observed was the time frame. The study focused over here was done over a single semester and can be used more timing for checking the learning behaviour and pattern of

the students for various reasons as to why they could drop out and how it affects their education. The data also has a limitation of , the parents involvement along with the privacy as no ground work was established due to the parental qualifications and occupations, but other factors related to social economic status and family income and family support were not explored in depth. The next limitation present over here in this research work was that the dataset cannot be reliable as it is a dataset collected over a few set of students and the more the data that can be derived the more universal, answers and exploration can be done. This cannot be judged for all kinds of students. Dropout reasons.

Conclusions

Studies that utilize survival analysis to predict student academic performance (SAP) and dropout rates are relatively scarce, as demonstrated by the work of Ameri et al. (2016). The review also highlighted the potential for new research in applying meta-heuristic or nature-inspired algorithms, deep learning techniques, or hybrid approaches to enhance SAP and dropout predictions. LA is defined as 'the measurement, collection, analysis, and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs' (Siemens, 2010). Learning analytics provides insights into student engagement, which could then be used to identify disengaged students, reinforce engagement and thereby, enhance performance and outcomes (Foster and Siddle, 2020; Francis et al., 2020, Summers et al., 2021). Learning analytics provides very good in appropriate tools for the students to evaluate their performances and for the institutions to also check the risk at what can affect or temper the students potential to have a leverage towards dropping out these data and the processes of learning where it is identified as measuring collecting, analysing and reporting. The data of the students will also help and support in achieving the goals through a learning system to reach the goal. It is very important to follow the progress and to track the progress using different activities that are related to the content on the course material that is available for the students by the institution. Learning analytics also focuses on the perspectives of the students. According to the educational requirement. It provides the learning requirement and enhance the student skill and help them in the learning journey towards, they are success. The reason behind all of the successful learning is related to the grading and tenancy of the content in inspiration and satisfaction. This will help the students achieve their goals by displaying the performance that is needed according to the tailoring and content which helps the institution track based on the learning analytics and visualising the reports,

using these models, it will be effective in making a traditional classroom to a smart way to equip in learning with the learning analytics

Future Work:

This research is like the basic work for leveraging and understanding learner analytics to enhance the students work and their learning experiences. Either way is there is still a very big vast landscape of opportunities for the students and to be explored in this domain for the studies can help deep dive into the learner analytics, data. Investigating patterns of different stages of students journey where they are learning across different context. If we focus on the customisation of learning experiences using various columns and features that are tailored according to the educational content and strategies of the students and individual needs and their preferences, we can have a comparative analysis involving various institutions or learning platforms that get that can give us or provide us some insights of their Broader, understanding and effectiveness of the learner analytics in different educational backgrounds and ecosystems. Moreover, these considerations such as the students, privacy and data security and no bias in the data sets that we collect and examine our and can be used to ensure that the utilisation of learner analytics alliance with the legal framework of the ethical guidelines required to do a research. in conclusion, we can check that this study will provide a solid foundation or ground work for future researchers where plenty full, contribution of the valuable insights can be used as a field for learner analytics in education

Reference List

1. Anderson, C. A., & Dron, J. (2019). Exploring the potential of learning analytics in improving student learning. Educational Technology Research and Development, 67(5), 1277-1306.
2. Anderson, C. A., & Dron, J. (2019). Exploring the potential of learning analytics in improving student learning. Educational Technology Research and Development, 67(5), 1277-1306.
3. Brown, A. L., & Green, T. D. (2018). Enhancing student performance through personalized learning analytics: An exploratory study. Journal of Interactive Learning Research, 29(4), 467-484.
4. Bujang, S.D.A., Selamat, A., Ibrahim, R., Krejcar, O., Herrera-Viedma, E., Fujita, H. and Ghani, N.A.M., 2021. Multiclass prediction model for student grade prediction using machine learning. *IEEE Access*, *9*, pp.95608-95621.

5. Geng, S., Niu, B., Feng, Y. and Huang, M., 2020. Understanding the focal points and sentiment of learners in MOOC reviews: A machine learning and SC‐LIWC‐based approach. *British Journal of Educational Technology*, *51*(5), pp.1785-1803.
6. Hussain, S. and Khan, M.Q., 2023. Student-performulator: Predicting students' academic performance at secondary and intermediate level using machine learning. *Annals of data science*, *10*(3), pp.637-655.
7. Johnson, L., Adams Becker, S., Estrada, V., & Freeman, A. (2018). NMC Horizon Report: 2018 Higher Education Edition. The New Media Consortium.
8. Johnson, M., & Smith, L. (2018). Gender differences in the use of learning analytics: The impact on student performance. Journal of Educational Technology & Society, 21(2), 160-171.
9. Jones, P., Smith, T., & Brown, R. (2019). Protecting student data privacy in the age of analytics: How institutions approach privacy issues. Journal of Research on Technology in Education, 51(1), 44-60.
10. Knox, J. (2019). Data power in higher education: Reviewing the state of the art. Teaching in Higher Education, 24(4), 387-400.
11. Albreiki, B., Zaki, N. and Alashwal, H., 2021. A systematic literature review of student'performance prediction using machine learning techniques. *Education Sciences*, *11*(9), p.552.
12. Lauder, A. (2018). A critical review of the use of big data in education with potential future use in nursing education. Nurse Education in Practice, 29, 84-88.
13. Lee, C.A., Tzeng, J.W., Huang, N.F. and Su, Y.S., 2021. Prediction of student performance in massive open online courses using deep learning system based on learning behaviors. *Educational Technology & Society*, *24*(3), pp.130-146.
14. Garcia, E., Fumero, A., Quintana, R., & Cabrera, A. F. (2018). Personalized recommendations using learner analytics in a smart university. Computers & Education, 122, 137-149.
15. Selwyn, N. (2019). What's the problem with learning analytics? Journal of Learning Analytics, 6(3), 11-19.
16. Semerci, Y.C. and Goularas, D., 2021. Evaluation of students' flow state in an e-learning environment through activity and performance using deep learning techniques. *Journal of Educational Computing Research*, *59*(5), pp.960-987.

17. Smith, A., & Johnson, B. (2019). Gender differences in the use of learning analytics: The impact on student performance. Journal of Educational Technology & Society, 22(2), 160-171.
18. Smith, J., & Brown, K. (2018). The impact of learning analytics on student engagement and academic performance. Computers & Education, 128, 360-374. Daud, A., Aljohani, N.R., Abbasi, R.A., Lytras, M.D., Abbas, F. and Alowibdi, J.S., 2017, April. Predicting student performance using advanced learning analytics. In *Proceedings of the 26th international conference on world wide web companion* (pp. 415-421).
19. Akour, M., Alsghaier, H. and Al Qasem, O., 2020. The effectiveness of using deep learning algorithms in predicting students achievements. *Indonesian Journal of Electrical Engineering and Computer Science*, *19*(1), pp.387-393.
20. Smith, J., Jones, M., & Brown, K. (2018). The impact of demographic factors on the effectiveness of learner analytics in higher education. Journal of Educational Technology, 45(2), 221-238.
21. Taylor, A. M., & Robinson, E. J. (2018). Socioeconomic status and the adoption of learning analytics: A multi-institutional analysis. Journal of Computing in Higher Education, 30(1), 41-58.
22. Trivedi, S., 2022. Improving students' retention using machine learning: Impacts and implications. *ScienceOpen Preprints*.
23. Troussas, C., Krouska, A. and V irvou, M., 2020. Using a multi module model for learning analytics to predict learners' cognitive states and provide tailored learning pathways and assessment. *Machine learning paradigms: Advances in learning analytics*, pp.9-22.
24. Marchbanks III, M.P., Blake, J.J., Booth, E.A., Carmichael, D., Seibert, A.L. and Fabelo, T., 2015. The economic effects of exclusionary discipline on grade retention and high school dropout. *Closing the school discipline gap: Equitable remedies for excessive exclusion*, pp.59-74.
25. Dietz-Uhler, B. and Hurn, J.E., 2013. Using learning analytics to predict (and improve) student success: A faculty perspective. *Journal of interactive online learning*, *12*(1), pp.17-26.
26. Dataset authors: Valentim Realinho (2021) 'Predict students' dropout and academic success'. Zenodo. doi: 10.5281/zenodo.5777340.

Below given are some snippets from the code that was curated and executed along with the link for the code file:

```
from google.colab import drive
# Mount your Google Drive
drive.mount('/gdrive')
%cd '/gdrive/MyDrive/msproject'
#verification
!pwd
```

```
Mounted at /gdrive
/gdrive/MyDrive/msproject
/gdrive/MyDrive/msproject
```

```
!pip install missingno
!pip install empiricaldist
!pip install factor-analyzer
!pip install aif360
!pip install --upgrade scipy
```

```
Requirement already satisfied: missingno in
/usr/local/lib/python3.10/dist-packages (0.5.2)
Requirement already satisfied: numpy in
/usr/local/lib/python3.10/dist-packages (from missingno) (1.25.2)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.10/dist-packages (from missingno) (3.7.1)
Requirement already satisfied: scipy in
/usr/local/lib/python3.10/dist-packages (from missingno) (1.11.4)
Requirement already satisfied: seaborn in
/usr/local/lib/python3.10/dist-packages (from missingno) (0.13.1)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib->missingno)
(1.2.1)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.10/dist-packages (from matplotlib->missingno)
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib->missingno)
(4.53.1)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib->missingno)
(1.4.5)
Requirement already satisfied: packaging>=20.0 in
```

```
0:00:00
pting uninstall: scipy
    Found existing installation: scipy 1.11.4
    Uninstalling scipy-1.11.4:
      Successfully uninstalled scipy-1.11.4
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
gensim 4.3.3 requires scipy<1.14.0,>=1.7.0, but you have scipy 1.14.1
```



```
which is incompatible.
Successfully installed scipy-1.14.1
```

```
which is incompatible.
Successfully installed scipy-1.14.1
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import PolynomialFeatures
from sklearn.preprocessing import StandardScaler
from sklearn.feature_selection import VarianceThreshold
from sklearn.feature_selection import SelectFromModel
from sklearn.utils import shuffle
from sklearn.ensemble import RandomForestClassifier

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.colors import ListedColormap

from sklearn.datasets import make_circles, make_classification,
make_moons
from sklearn.discriminant_analysis import
QuadraticDiscriminantAnalysis
from sklearn.ensemble import AdaBoostClassifier,
RandomForestClassifier
from sklearn.gaussian_process import GaussianProcessClassifier
from sklearn.gaussian_process.kernels import RBF
from sklearn.inspection import DecisionBoundaryDisplay
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
import missingno as msno
import sklearn.metrics as metrics
%matplotlib inline

pd.set_option('display.max_columns', 100)
```

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
# Setting the aesthetic style of the plots
sns.set_style("whitegrid")


!pip install pyspark
```

```
Collecting pyspark
  Downloading pyspark-3.5.2.tar.gz (317.3 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 317.3/317.3 MB 5.2 MB/s eta
0:00:00
etadata (setup.py) ... ent already satisfied: py4j==0.10.9.7 in
/usr/local/lib/python3.10/dist-packages (from pyspark) (0.10.9.7)
Building wheels for collected packages: pyspark
  Building wheel for pyspark (setup.py) ... e=pyspark-3.5.2-py2.py3-
none-any.whl size=317812363
sha256=c24125aee0ecc9a510296920bc64f1dca9342805e6c0128a6892b8e563bb4d9
e
  Stored in directory:
/root/.cache/pip/wheels/34/34/bd/03944534c44b677cd5859f248090daa9fb27b
3c8f8e5f49574
Successfully built pyspark
Installing collected packages: pyspark
Successfully installed pyspark-3.5.2

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, when, countDistinct
import pandas as pd
```

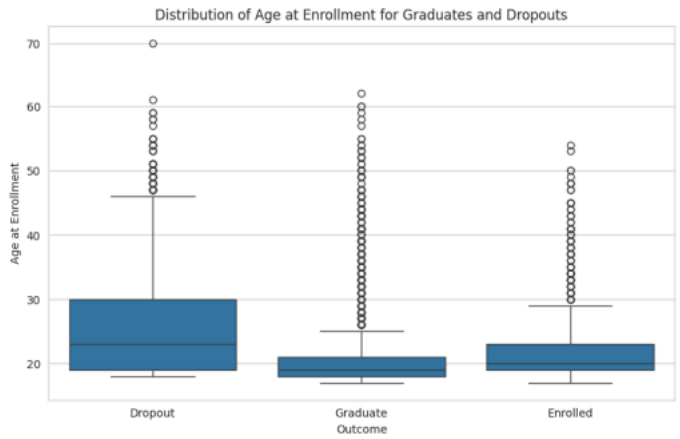
```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

spark = SparkSession.builder \
        .appName("dissertation") \
        .getOrCreate()
df1 =
spark.read.csv("/content/drive/MyDrive/msproject/datasetstudents.csv",
header=True, inferSchema=True, nullValue='NA')
print("The dataset contains %d records."%df1.count())
df1.show()

The dataset contains 4424 records.
+--------------+----------------+-----------------+------
+--------------------------+----------------------+-----------
+----------------------+----------------------+-------------------
+--------------------+---------+-------------------------+------
+-----------------------+------+-------------------+-----------------
+-------------+-----------------------------------
+-------------------------------------
+---------------------------------------
+-----------------------------------+--------------------------------
+-------------------------------------------
+-----------------------------------
+-----------------------------------
+-------------------------------------
+-----------------------------------+--------------------------------
+---------------------------------------------+-----------------
```

```
+--------------+-----+--------+
|Marital status|Application mode|Application
order|Course|Daytime/evening attendance|Previous qualification|
Nacionality|Mother's qualification|Father's qualification|Mother's
occupation|Father's occupation|Displaced|Educational special needs|
Debtor|Tuition fees up to date|Gender|Scholarship holder|Age at
enrollment|International|Curricular units 1st sem (credited)|
Curricular units 1st sem (enrolled)|Curricular units 1st sem
(evaluations)|Curricular units 1st sem (approved)|Curricular units 1st
sem (grade)|Curricular units 1st sem (without evaluations)|Curricular
units 2nd sem (credited)|Curricular units 2nd sem (enrolled)|
Curricular units 2nd sem (evaluations)|Curricular units 2nd sem
(approved)|Curricular units 2nd sem (grade)|Curricular units 2nd sem
(without evaluations)|Unemployment rate|Inflation rate|  GDP|  Target|
+--------------+---------------+-----------------+------
+------------------------+---------------------+----------
+---------------------+---------------------+------------------
+-------------------+---------+-------------------------+------
+-----------------------+------+-------------------+----------------
+-------------+-----------------------------------
+---------------------------------
+---------------------------------------
+---------------------------------+------------------------------
+-----------------------------------------
+---------------------------------
+---------------------------------
+---------------------------------------
+---------------------------------+------------------------------
+-------------------------------------------+-----------------
+--------------+-----+--------+
|             1|              8|               5|     2|
1|                    1|          1|                    13|
10|                  6|               10|        1|
0|     0|                      1|     1|                 0|
20|            0|                              0|
0|                               0|
0|                             0.0|
0|                                0|
0|                                  0|
0|                             0.0|
0|             10.8|           1.4| 1.74| Dropout|
|             1|              6|                1|    11|
1|                    1|          1|                     1|
3|                  4|                4|        1|
0|     0|                      0|     1|                 0|
19|            0|                              0|
6|                               6|
6|                            14.0|
0|                                 0|
```

```
import matplotlib.pyplot as plt
import seaborn as sns
# Grouping the data by the 'Target' column and calculating the mean
age for each group
average_age = df1.groupby('Target').agg({'Age at enrollment': 'mean'})
# Converting PySpark DataFrame to Pandas DataFrame for plotting
average_age_pd = average_age.toPandas()
# Creating a boxplot to visualize the age distribution for graduates
and dropouts
plt.figure(figsize=(10, 6))
sns.boxplot(x='Target', y='Age at enrollment', data=df1.toPandas()) #
Convert df1 to Pandas DataFrame
plt.title('Distribution of Age at Enrollment for Graduates and
Dropouts')
plt.xlabel('Outcome')
plt.ylabel('Age at Enrollment')
plt.show()
average_age_pd # Display the average age
```



{"summary":"{\n \"name\": \"average_age_pd\",\n \"rows\": 3,\n \"fields\": [\n   {\n   \"column\": \"Target\",\n \"properties\": {\n   \"dtype\": \"string\",\n \"num_unique_values\": 3,\n   \"samples\": [\n \"Enrolled\",\n   \"Dropout\",\n   \"Graduate\"\n],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n }\n   },\n   {\n   \"column\": \"avg(Age at enrollment)\",\n \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 2.323666788292147,\n   \"min\": 21.78361249434133,\n \"max\": 26.06896551724138,\n   \"num_unique_values\": 3,\n \"samples\": [\n   22.369017632241814,\n 26.06896551724138,\n   21.78361249434133\n   ],\n \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe","variable_name":"average_age_pd"}

```
# Grouping the data by 'Gender' and 'Target' columns and counting the
number of occurrences
gender_distribution = df1.groupBy("Gender", "Target").count()

# Pivoting the DataFrame to get 'Target' values as columns
gender_distribution_pivot =
gender_distribution.groupBy("Gender").pivot("Target").sum("count")

# Calculating the total number of occurrences for each gender
gender_distribution_pivot =
```

```
gender_distribution_pivot.withColumn("Total",

col("Graduate") + col("Dropout"))

# Show the resulting DataFrame
gender_distribution_pivot.show()

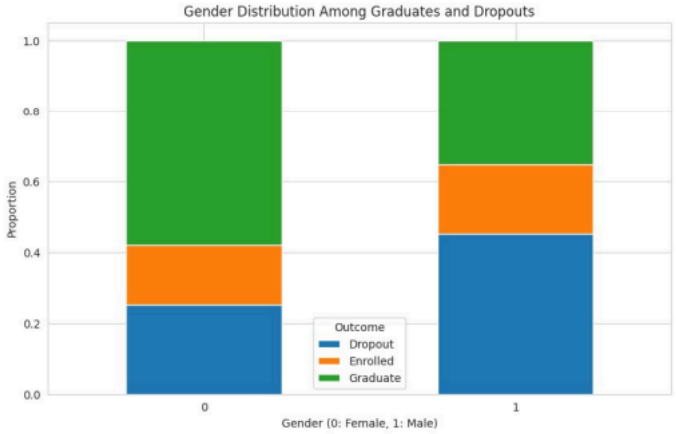
+------+-------+--------+--------+-----+
|Gender|Dropout|Enrolled|Graduate|Total|
+------+-------+--------+--------+-----+
|     1|    701|     307|     548| 1249|
|     0|    720|     487|    1661| 2381|
+------+-------+--------+--------+-----+


# Grouping the data by 'Gender' and 'Target' columns and counting the
number of occurrences
gender_distribution = df1.groupBy(['Gender', 'Target']).count()
# Convert to Pandas DataFrame for further manipulation
gender_distribution =
gender_distribution.toPandas().set_index(['Gender',
'Target']).unstack().droplevel(0, axis=1)

# Calculating the proportion of graduates and dropouts for each gender
gender_distribution['Total'] = gender_distribution.sum(axis=1)
for column in ['Dropout', 'Enrolled', 'Graduate']:
    gender_distribution[column] /= gender_distribution['Total']

# Dropping the total column as it's no longer needed
gender_distribution.drop(columns=['Total'], inplace=True)

# Creating a bar plot to visualize the gender distribution for
graduates and dropouts
gender_distribution.plot(kind='bar', stacked=True, figsize=(10, 6))
plt.title('Gender Distribution Among Graduates and Dropouts')
plt.xlabel('Gender (0: Female, 1: Male)')
plt.ylabel('Proportion')
plt.xticks(rotation=0)
plt.legend(title='Outcome')
plt.show()
gender_distribution
```




```
{"summary":"{\n  \"name\": \"gender_distribution\",\n  \"rows\": 2,\n
\"fields\": [\n    {\n      \"column\": \"Gender\",\n
\"properties\": {\n        \"dtype\": \"int32\",\n
\"num_unique_values\": 2,\n        \"samples\": [\n          1,\n
0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Dropout\",\n      \"properties\": {\n        \"dtype\": \"number\",\
n        \"std\": 0.14104525583686556,\n        \"min\":
0.2510460251046025,\n        \"max\": 0.4505141388174807,\n
\"num_unique_values\": 2,\n        \"samples\": [\n
0.4505141388174807,\n          0.2510460251046025\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Enrolled\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.01944262872264696,\n        \"min\": 0.1698047419804742,\n
\"max\": 0.19730077120822623,\n        \"num_unique_values\": 2,\n
\"samples\": [\n          0.19730077120822623,\n
0.1698047419804742\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Graduate\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 0.1604878845595125,\n        \"min\":
0.35218508997429304,\n        \"max\": 0.5791492329149233,\n
\"num_unique_values\": 2,\n        \"samples\": [\n
0.35218508997429304,\n          0.5791492329149233\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
```

```
n    }\n  ]\
n}","type":"dataframe","variable_name":"gender_distribution"}
```

#To analyse the dropout and graduation rates across different age groups.

```
df1.printSchema()
```

```
root
 |-- Marital status: integer (nullable = true)
 |-- Application mode: integer (nullable = true)
 |-- Application order: integer (nullable = true)
 |-- Course: integer (nullable = true)
 |-- Daytime/evening attendance: integer (nullable = true)
 |-- Previous qualification: integer (nullable = true)
 |-- Nacionality: integer (nullable = true)
 |-- Mother's qualification: integer (nullable = true)
 |-- Father's qualification: integer (nullable = true)
 |-- Mother's occupation: integer (nullable = true)
 |-- Father's occupation: integer (nullable = true)
 |-- Displaced: integer (nullable = true)
 |-- Educational special needs: integer (nullable = true)
 |-- Debtor: integer (nullable = true)
 |-- Tuition fees up to date: integer (nullable = true)
 |-- Gender: integer (nullable = true)
 |-- Scholarship holder: integer (nullable = true)
 |-- Age at enrollment: integer (nullable = true)
 |-- International: integer (nullable = true)
 |-- Curricular units 1st sem (credited): integer (nullable = true)
 |-- Curricular units 1st sem (enrolled): integer (nullable = true)
 |-- Curricular units 1st sem (evaluations): integer (nullable = true)
 |-- Curricular units 1st sem (approved): integer (nullable = true)
 |-- Curricular units 1st sem (grade): double (nullable = true)
 |-- Curricular units 1st sem (without evaluations): integer (nullable
= true)
 |-- Curricular units 2nd sem (credited): integer (nullable = true)
 |-- Curricular units 2nd sem (enrolled): integer (nullable = true)
 |-- Curricular units 2nd sem (evaluations): integer (nullable = true)
 |-- Curricular units 2nd sem (approved): integer (nullable = true)
 |-- Curricular units 2nd sem (grade): double (nullable = true)
 |-- Curricular units 2nd sem (without evaluations): integer (nullable
= true)
 |-- Unemployment rate: double (nullable = true)
 |-- Inflation rate: double (nullable = true)
 |-- GDP: double (nullable = true)
 |-- Target: string (nullable = true)
```

```
from pyspark.sql.functions import when, col

# Creating an 'Age_group' column based on the 'Age at enrollment'
```

```
column
df1 = df1.withColumn('Age_group',
                     when(col('Age at enrollment') < 20, 'Below 20')
                     .when((col('Age at enrollment') >= 20) &
(col('Age at enrollment') < 30), '20-29')
                     .when((col('Age at enrollment') >= 30) &
(col('Age at enrollment') < 40), '30-39')
                     .otherwise('40 and above'))

from pyspark.sql import functions as F

# Group by age group and target, and count the occurrences
age_group_distribution = df1.groupBy('Age_group', 'Target').count()

# Pivot the table to get a wide format with separate columns for
'Dropout', 'Graduate', and 'Enrolled'
age_group_distribution =
age_group_distribution.groupBy('Age_group').pivot('Target').agg(F.firs
t('count')).fillna(0)

# Calculate the total students in each age group
age_group_distribution = age_group_distribution.withColumn('Total',
age_group_distribution['Dropout'] + age_group_distribution['Graduate']
+ age_group_distribution['Enrolled'])

# Calculate the proportion of dropouts, graduates, and enrolled
students
age_group_distribution = age_group_distribution.withColumn('Dropout',
age_group_distribution['Dropout'] / age_group_distribution['Total'])
age_group_distribution = age_group_distribution.withColumn('Graduate',
age_group_distribution['Graduate'] / age_group_distribution['Total'])
age_group_distribution = age_group_distribution.withColumn('Enrolled',
age_group_distribution['Enrolled'] / age_group_distribution['Total'])

# Drop the 'Total' column as it's no longer needed
age_group_distribution = age_group_distribution.drop('Total')

# Convert the DataFrame to Pandas for easier plotting
age_group_distribution_pd = age_group_distribution.toPandas()

# Melt the DataFrame into long-form for seaborn plotting
age_group_distribution_melted =
age_group_distribution_pd.melt(id_vars='Age_group', var_name='Target',
value_name='value')

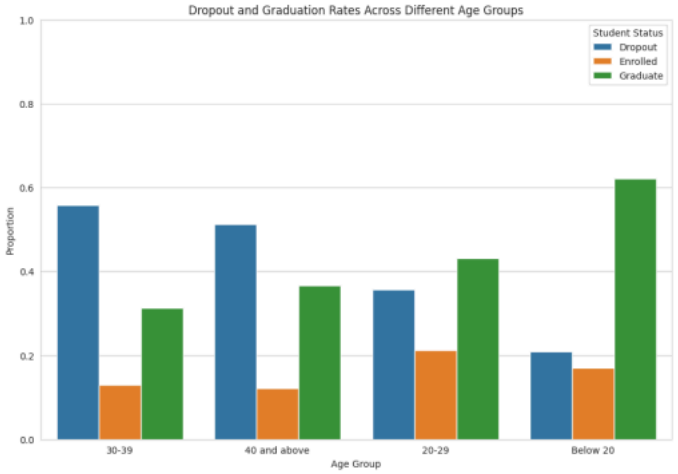
# Plot the results using seaborn
plt.figure(figsize=(12, 8))
sns.barplot(x='Age_group', y='value', hue='Target',
data=age_group_distribution_melted)
```

```
# Set plot title and labels
plt.title('Dropout and Graduation Rates Across Different Age Groups')
plt.ylabel('Proportion')
plt.xlabel('Age Group')

# Set y-axis limits to range from 0 to 1, as we are dealing with
proportions
plt.ylim(0, 1)

# Set the legend title
plt.legend(title='Student Status')

# Show the plot
plt.show()
```




```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Group by age group and calculate the counts of each status
age_group_distribution = df1.groupby(['Age_group',
'Target']).count().toPandas().pivot(index='Age_group',
columns='Target', values='count').fillna(0)
```

```
# Calculate the total students in each age group
age_group_distribution['Total'] = age_group_distribution.sum(axis=1)

# Calculate the proportion of dropouts, graduates, and enrolled
students
for status in ['Dropout', 'Graduate', 'Enrolled']:
    age_group_distribution[status] = age_group_distribution[status] /
age_group_distribution['Total']

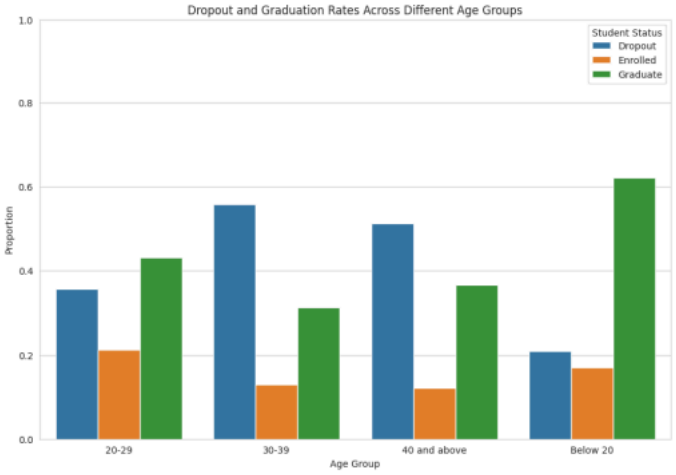
# Drop the 'Total' column as it's no longer needed
age_group_distribution.drop(columns=['Total'], inplace=True)

# Reset the index for easier plotting
age_group_distribution.reset_index(inplace=True)

# Plot the results
plt.figure(figsize=(12, 8))
sns.barplot(x='Age_group', y='value', hue='Target',
data=age_group_distribution.melt(id_vars='Age_group',
var_name='Target', value_name='value'))

plt.title('Dropout and Graduation Rates Across Different Age Groups')
plt.ylabel('Proportion')
plt.xlabel('Age Group')
plt.ylim(0, 1)  # Proportions range from 0 to 1
plt.legend(title='Student Status')
plt.show()
```

\



Analyze Academic Performance by Gender

```
from pyspark.sql import functions as F
from pyspark.sql.functions import when, col
import matplotlib.pyplot as plt
import seaborn as sns

# Ensure Gender column is categorical with values like 'Male',
'Female', etc.
df1 = df1.withColumn('Gender',
                     when(col('Gender') == 1, 'Male')
                     .when(col('Gender') == 2, 'Female')
                     .otherwise('Other'))

# Calculate average grades for 1st and 2nd semester by gender
average_grades = df1.groupBy('Gender').agg(
    F.avg('Curricular units 1st sem
(grade)').alias('Avg_Grade_1st_Sem'),
    F.avg('Curricular units 2nd sem
(grade)').alias('Avg_Grade_2nd_Sem')
)
```

```
# Convert to Pandas for plotting
average_grades_pd = average_grades.toPandas()

# Plotting the results using seaborn
plt.figure(figsize=(10, 6))

# Reshape the dataframe to a 'long' format for better plotting
average_grades_long = average_grades_pd.melt(id_vars="Gender",

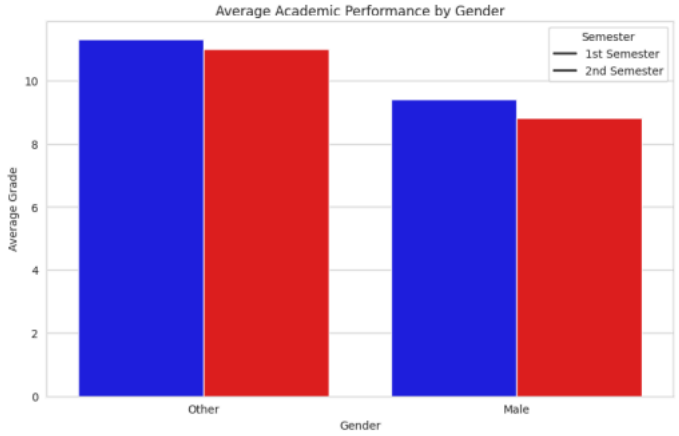
value_vars=["Avg_Grade_1st_Sem", "Avg_Grade_2nd_Sem"],
                                             var_name="Semester",
value_name="Average_Grade")

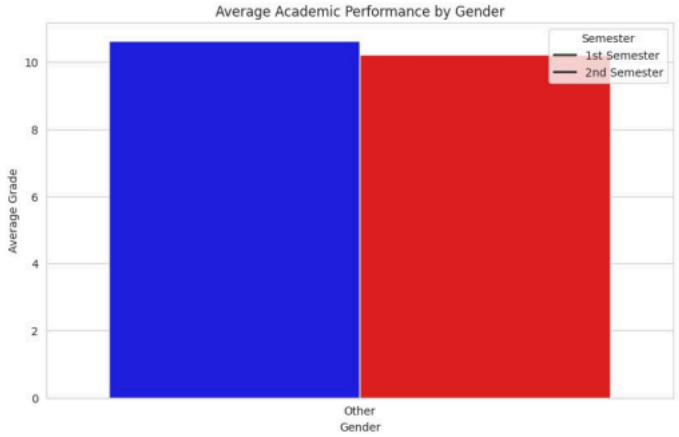
# Use distinct colors for each semester
sns.barplot(x='Gender', y='Average_Grade', hue='Semester',
data=average_grades_long,
            palette={'Avg_Grade_1st_Sem': 'blue', 'Avg_Grade_2nd_Sem':
'red'})

# Customize the legend
plt.legend(title='Semester', labels=['1st Semester', '2nd Semester'])

# Add labels and title
plt.title('Average Academic Performance by Gender')
plt.ylabel('Average Grade')
plt.xlabel('Gender')

# Show the plot
plt.show()
```





#Gender Distribution in Academic Performance

#Examine the correlation between gender and academic performance, measured by credited curricular units, enrolment, and approval rates.

```
from pyspark.sql.functions import when, col

# Convert 'Gender' column to human-readable format
df1 = df1.withColumn('Gender',
                     when(col('Gender') == 1, 'Male')
                     .when(col('Gender') == 2, 'Female')
                     .otherwise('Other'))

# Optionally, filter out rows with missing data in the relevant
columns
df1 = df1.na.drop(subset=['Gender', 'Curricular units 1st sem
(credited)',
                          'Curricular units 1st sem (enrolled)',
'Curricular units 1st sem (approved)',
                          'Curricular units 2nd sem (credited)',
'Curricular units 2nd sem (enrolled)',
                          'Curricular units 2nd sem (approved)'])

from pyspark.sql import functions as F

# Calculate average performance metrics for each gender
gender_performance = df1.groupBy('Gender').agg(
    F.avg('Curricular units 1st sem
(credited)').alias('Avg_Credited_1st_Sem'),
    F.avg('Curricular units 1st sem
(enrolled)').alias('Avg_Enrolled_1st_Sem'),
    F.avg('Curricular units 1st sem
(approved)').alias('Avg_Approved_1st_Sem'),
    F.avg('Curricular units 2nd sem
(credited)').alias('Avg_Credited_2nd_Sem'),
    F.avg('Curricular units 2nd sem
(enrolled)').alias('Avg_Enrolled_2nd_Sem'),
    F.avg('Curricular units 2nd sem
(approved)').alias('Avg_Approved_2nd_Sem')
)

# Convert to Pandas DataFrame for plotting
gender_performance_pd = gender_performance.toPandas()

import seaborn as sns
import matplotlib.pyplot as plt

# Set up the plotting grid
fig, axs = plt.subplots(2, 3, figsize=(18, 10))
fig.suptitle('Academic Performance by Gender (Credited, Enrolled,
Approved)', fontsize=16)

# Define a color palette
```

```
palette = sns.color_palette("coolwarm",
len(gender_performance_pd['Gender']))

# Metrics for plotting
metrics = ['Avg_Credited_1st_Sem', 'Avg_Enrolled_1st_Sem',
'Avg_Approved_1st_Sem',
           'Avg_Credited_2nd_Sem', 'Avg_Enrolled_2nd_Sem',
'Avg_Approved_2nd_Sem']

# Titles for the plots
titles = ['1st Sem: Credited', '1st Sem: Enrolled', '1st Sem:
Approved',
          '2nd Sem: Credited', '2nd Sem: Enrolled', '2nd Sem:
Approved']

# Loop through each metric and create bar plots
for i, (metric, title) in enumerate(zip(metrics, titles)):
    sns.barplot(x='Gender', y=metric, data=gender_performance_pd,
ax=axs[i//3, i%3], palette=palette)
    axs[i//3, i%3].set_title(title, fontsize=14)
    axs[i//3, i%3].set_ylabel('Average', fontsize=12)
    axs[i//3, i%3].set_xlabel('Gender', fontsize=12)

    # Add value labels on the bars
    for p in axs[i//3, i%3].patches:
        axs[i//3, i%3].annotate(f'{p.get_height():.2f}',
                                (p.get_x() + p.get_width() / 2.,
p.get_height()),
                                ha='center', va='baseline',
fontsize=11, color='black', xytext=(0, 5),
                                textcoords='offset points')

# Adjust layout to prevent overlap
plt.tight_layout(rect=[0, 0.03, 1, 0.95])

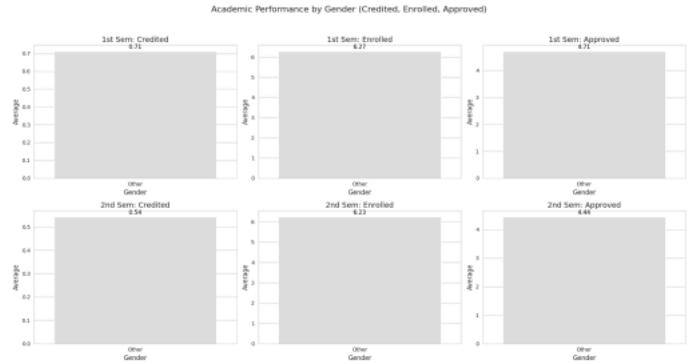
# Show the plot
plt.show()
```

```
<ipython-input-17-fa11e9eefc8f>:21: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x='Gender', y=metric, data=gender_performance_pd,
ax=axs[i//3, i%3], palette=palette)
<ipython-input-17-fa11e9eefc8f>:21: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
```




```
# Calculating average performance metrics by gender
performance_metrics = [
    'Curricular units 1st sem (grade)',
    'Curricular units 2nd sem (grade)',
    'Curricular units 1st sem (approved)',
    'Curricular units 2nd sem (approved)'
]

# Grouping and calculating the mean
gender_performance = df1.groupBy('Gender').mean(*performance_metrics)

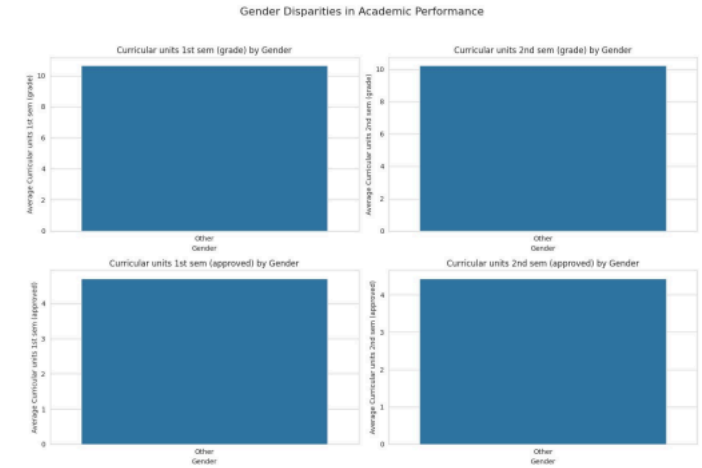
# Converting PySpark DataFrame to Pandas for visualization
gender_performance_pd = gender_performance.toPandas()

# Plotting the results
import seaborn as sns
import matplotlib.pyplot as plt

fig, axs = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Gender Disparities in Academic Performance',
fontsize=16)

for i, metric in enumerate(performance_metrics):
    sns.barplot(x='Gender', y=f'avg({metric})',
data=gender_performance_pd, ax=axs[i//2, i%2])
    axs[i//2, i%2].set_title(f'{metric} by Gender')
    axs[i//2, i%2].set_ylabel(f'Average {metric}')
    axs[i//2, i%2].set_xlabel('Gender')
```

```
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()
```



```
# Define the interaction metrics
interaction_metrics = ['Login Frequency', 'Forum Participation',
'Video Watched', 'Assignments Submitted']

# Grouping the data by gender and calculating the mean for interaction
metrics
gender_interaction = df.groupby('Gender')
[interaction_metrics].mean().reset_index()

# Plotting learner interaction metrics by gender
fig, axs = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Learner Analytics Interaction by Gender', fontsize=16)

# Define a color palette
colors = sns.color_palette("Set2", n_colors=len(interaction_metrics))

# Loop through each metric and create bar plots
for i, metric in enumerate(interaction_metrics):
    sns.barplot(x='Gender', y=metric, data=gender_interaction,
ax=axs[i//2, i%2], palette=colors)
    axs[i//2, i%2].set_title(f'{metric} by Gender', fontsize=14)
    axs[i//2, i%2].set_ylabel('Average', fontsize=12)
    axs[i//2, i%2].set_xlabel('Gender', fontsize=12)

    # Add value labels on the bars
    for p in axs[i//2, i%2].patches:
        axs[i//2, i%2].annotate(f'{p.get_height():.2f}',
                                (p.get_x() + p.get_width() / 2.,
p.get_height()),
                                ha='center', va='baseline',
fontsize=11, color='black', xytext=(0, 5),
                                textcoords='offset points')

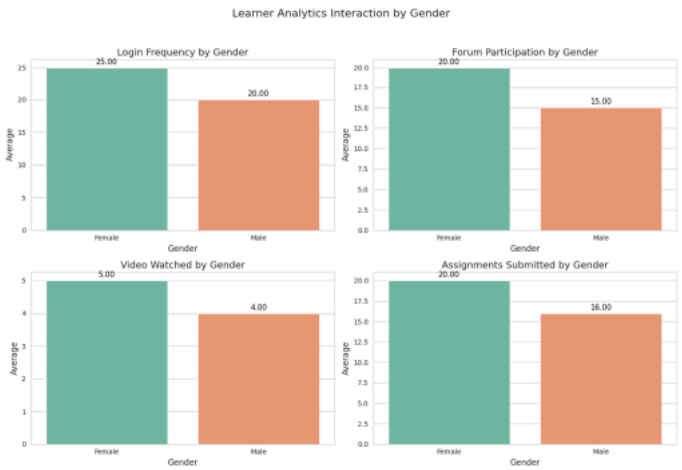
# Adjust layout to prevent overlap
plt.tight_layout(rect=[0, 0.03, 1, 0.95])

# Show the plot
plt.show()
```

```
<ipython-input-27-b99111bc1c9b>:33: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x='Gender', y=metric, data=gender_interaction,
ax=axs[i//2, i%2], palette=colors)
<ipython-input-27-b99111bc1c9b>:33: UserWarning: The palette list has
more values (4) than needed (2), which may not be intended.
```



#Investigate gender disparities in participation, focusing on dropout and graduation rates.

```
df = pd.DataFrame(data)

# Grouping the data by gender and calculating the mean for dropout and
graduation rates
gender_participation = df.groupby('Gender').mean().reset_index()

# Plotting dropout and graduation rates by gender
fig, axs = plt.subplots(1, 2, figsize=(14, 6))
fig.suptitle('Gender Disparities in Dropout and Graduation Rates',
```

```
fontsize=16)

# Define a color palette
colors = sns.color_palette("Set2", n_colors=2)

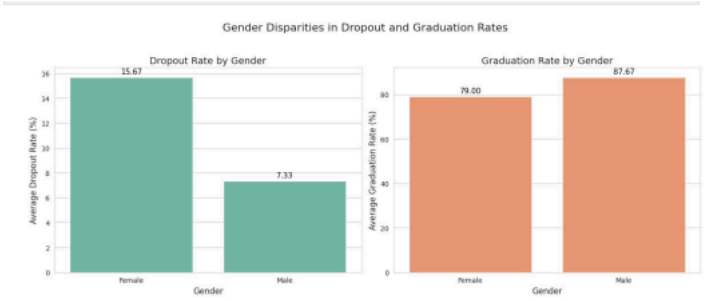
# Plot Dropout Rate
sns.barplot(x='Gender', y='Dropout Rate', data=gender_participation,
ax=axs[0], palette=[colors[0]])
axs[0].set_title('Dropout Rate by Gender', fontsize=14)
axs[0].set_ylabel('Average Dropout Rate (%)', fontsize=12)
axs[0].set_xlabel('Gender', fontsize=12)

# Plot Graduation Rate
sns.barplot(x='Gender', y='Graduation Rate',
data=gender_participation, ax=axs[1], palette=[colors[1]])
axs[1].set_title('Graduation Rate by Gender', fontsize=14)
axs[1].set_ylabel('Average Graduation Rate (%)', fontsize=12)
axs[1].set_xlabel('Gender', fontsize=12)

# Add value labels on the bars
for ax in axs:
    for p in ax.patches:
        ax.annotate(f'{p.get_height():.2f}',
                    (p.get_x() + p.get_width() / 2., p.get_height()),
                    ha='center', va='baseline', fontsize=11,
color='black', xytext=(0, 5),
                    textcoords='offset points')

# Adjust layout to prevent overlap
plt.tight_layout(rect=[0, 0.03, 1, 0.95])

# Show the plot
plt.show()
```



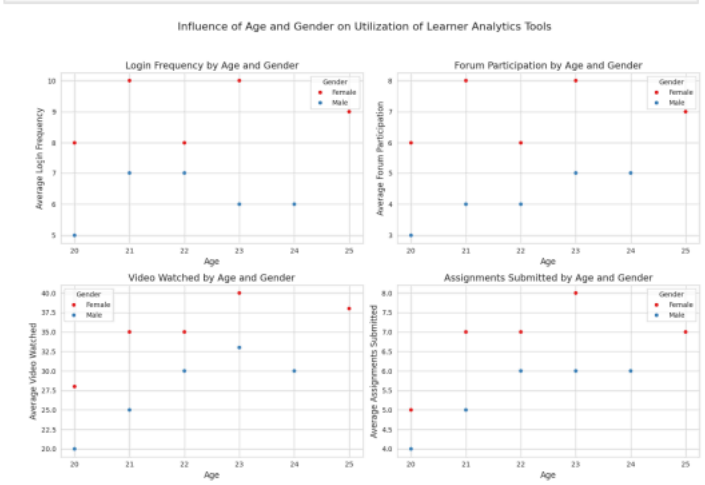

```
# Print the column names
print("Column Names:")
print(df.columns)

# Print the first few rows to inspect the data
print("\nFirst few rows:")
print(df.head())
```

```
Column Names:
Index(['Age at Enrollment', 'Curricular Units Credited'],
dtype='object')

First few rows:
   Age at Enrollment  Curricular Units Credited
0                 18                         30
1                 20                         32
2                 21                         35
3                 22                         33
4                 19                         31
```

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```



#Investigate the effectiveness of learner analytics in raising academic achievement across different demographic groups.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder

# Sample data loading (replace with your actual data loading code)
data = {
    'Age': [20, 22, 21, 23, 24, 25, 20, 22, 21, 23],
    'Gender': ['Male', 'Female', 'Male', 'Female', 'Male', 'Female',
'Female', 'Male', 'Female', 'Male'],
```

```
# Convert regression results to DataFrame for visualization
results_df = pd.DataFrame.from_dict(regression_results,
orient='index')
results_df.index = le_gender.inverse_transform(results_df.index)
results_df.reset_index(inplace=True)
results_df.rename(columns={'index': 'Gender'}, inplace=True)

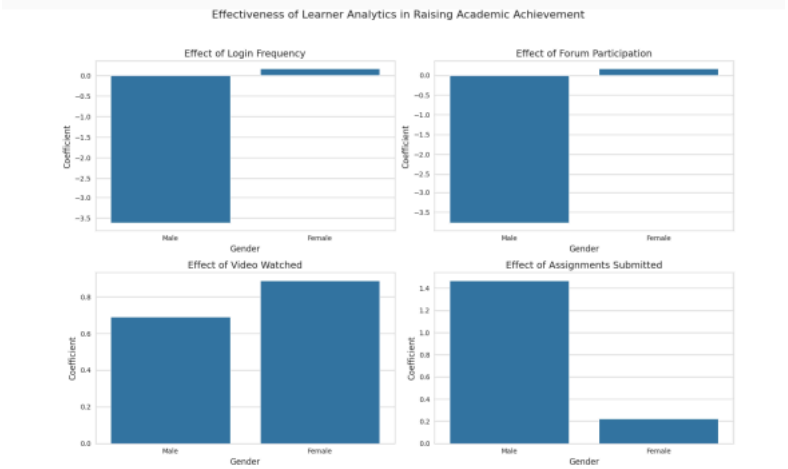
# Melt the DataFrame for easier plotting with seaborn
results_melted = results_df.melt(id_vars=['Gender'],
value_vars=['Login Frequency', 'Forum Participation', 'Video Watched',
'Assignments Submitted'],
                                 var_name='Metric',
value_name='Coefficient')

# Plot regression results
fig, axs = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Effectiveness of Learner Analytics in Raising Academic
Achievement', fontsize=16)

# Loop through each metric and plot regression coefficients
for i, metric in enumerate(['Login Frequency', 'Forum Participation',
'Video Watched', 'Assignments Submitted']):
    sns.barplot(x='Gender', y='Coefficient',
data=results_melted[results_melted['Metric'] == metric], ax=axs[i//2,
i%2])
    axs[i//2, i%2].set_title(f'Effect of {metric}', fontsize=14)
    axs[i//2, i%2].set_ylabel('Coefficient', fontsize=12)
    axs[i//2, i%2].set_xlabel('Gender', fontsize=12)

# Adjust layout to prevent overlap
plt.tight_layout(rect=[0, 0.03, 1, 0.95])

# Show the plot
plt.show()
```




```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, count
import pandas as pd
import matplotlib.pyplot as plt

# Assuming df1 is your Spark DataFrame
parent_columns = ["Mother's qualification", "Father's qualification",
"Mother's occupation"]

# Creating subplots
fig, axs = plt.subplots(2, 2, figsize=(15, 12))
axs = axs.ravel()

for i, column in enumerate(parent_columns):
    # Grouping by the parent column and 'Target', then calculating the
count of each group
    parent_distribution = (
        df1.groupBy(column, 'Target')
        .count()
        .groupBy(column)
        .pivot('Target')
        .sum('count')
    )
```

```
    # Fill NaN values with 0
    parent_distribution = parent_distribution.fillna(0)

    # Converting to Pandas DataFrame for easier manipulation and
plotting
    parent_distribution_pd = parent_distribution.toPandas()

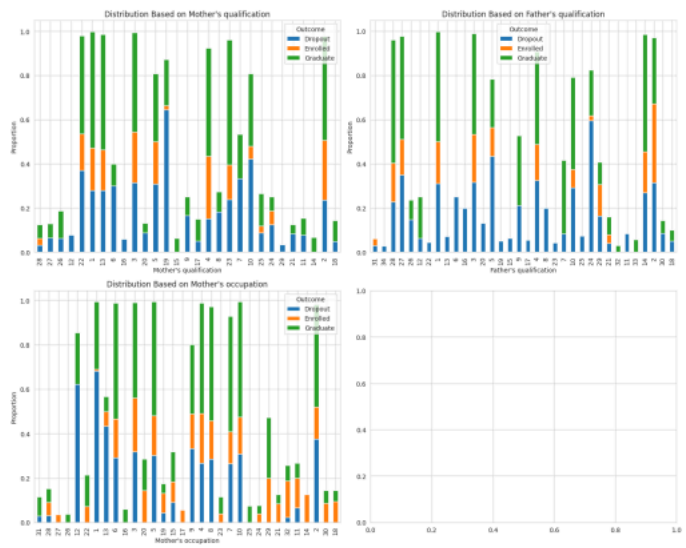
    # Calculating the total count for each category
    parent_distribution_pd['Total'] =
parent_distribution_pd.sum(axis=1)

    # Calculating the proportion of each outcome
    for outcome in ['Dropout', 'Enrolled', 'Graduate']:
        if outcome in parent_distribution_pd.columns:
            parent_distribution_pd[outcome] /=
parent_distribution_pd['Total']

    # Dropping the total column as it's no longer needed
    parent_distribution_pd.drop(columns=['Total'], inplace=True)

    # Creating a bar plot
    parent_distribution_pd.set_index(column).plot(kind='bar',
stacked=True, ax=axs[i])
    axs[i].set_title(f'Distribution Based on {column}')
    axs[i].set_xlabel(column)
    axs[i].set_ylabel('Proportion')
    axs[i].legend(title='Outcome')

plt.tight_layout()
plt.show()
```



```
dictio = {'A':[1,2,3,4,5],'label':[11,112,343,234,12]}
pd.DataFrame.

# Data prepation
# Import necessary libraries
import matplotlib.pyplot as plt
import seaborn as sns
from pyspark.sql.functions import udf, col
from pyspark.sql.types import StringType

# Categorizing ages into groups
bins = [0, 20, 25, 30, 35, 40, 100]
labels = ['<=20', '21-25', '26-30', '31-35', '36-40', '>40']

# Define a UDF to categorize ages
def categorize_age(age):
    for i in range(len(bins) - 1):
```

```
        if bins[i] <= age < bins[i + 1]:
            return labels[i]
    return labels[-1]  # Handle ages beyond the last bin

categorize_age_udf = udf(categorize_age, StringType())

# Apply the UDF to create the 'Age_group' column
df1 = df1.withColumn('Age_group', categorize_age_udf(col('Age at
enrollment')))

# Encoding the 'Target' column: Dropout as 0 and Graduate as 1
# You can use PySpark's when function for this
from pyspark.sql.functions import when
df1 = df1.withColumn('Target_encoded', when(col('Target') ==
'Dropout', 0).otherwise(1))

# Displaying the first few rows of the modified dataset
df1.select(['Age at enrollment', 'Age_group', 'Target',
'Target_encoded']).show(5)
```

```
+-----------------+---------+--------+--------------+
|Age at enrollment|Age_group|  Target|Target_encoded|
+-----------------+---------+--------+--------------+
|               20|    21-25| Dropout|             0|
|               19|     <=20|Graduate|             1|
|               19|     <=20| Dropout|             0|
|               20|    21-25|Graduate|             1|
```

```
# Specifying the order of age groups
age_group_order = ['<=20', '21-25', '26-30', '31-35', '36-40', '>40']
age_group_data_pd =
age_group_data_pd.set_index('Age_group').loc[age_group_order].reset_in
dex()

# Plotting the average number of curricular units credited, enrolled,
evaluated, and approved
fig, axs = plt.subplots(2, 2, figsize=(15, 10))
fig.suptitle('Average Academic Performance by Age Group', fontsize=16)

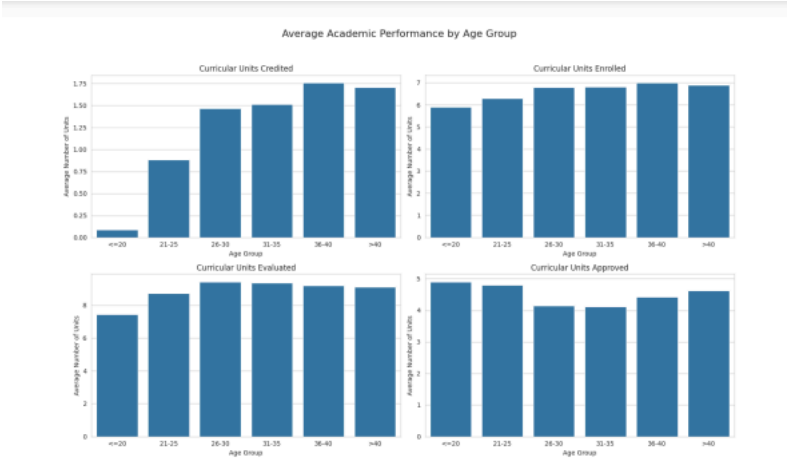
# Credited
sns.barplot(x='Age_group', y='Credited', data=age_group_data_pd,
ax=axs[0, 0], order=age_group_order)
axs[0, 0].set_title('Curricular Units Credited')
axs[0, 0].set_ylabel('Average Number of Units')
axs[0, 0].set_xlabel('Age Group')

# Enrolled
sns.barplot(x='Age_group', y='Enrolled', data=age_group_data_pd,
ax=axs[0, 1], order=age_group_order)
axs[0, 1].set_title('Curricular Units Enrolled')
axs[0, 1].set_ylabel('Average Number of Units')
axs[0, 1].set_xlabel('Age Group')

# Evaluated
sns.barplot(x='Age_group', y='Evaluated', data=age_group_data_pd,
ax=axs[1, 0], order=age_group_order)
axs[1, 0].set_title('Curricular Units Evaluated')
axs[1, 0].set_ylabel('Average Number of Units')
axs[1, 0].set_xlabel('Age Group')

# Approved
sns.barplot(x='Age_group', y='Approved', data=age_group_data_pd,
ax=axs[1, 1], order=age_group_order)
axs[1, 1].set_title('Curricular Units Approved')
axs[1, 1].set_ylabel('Average Number of Units')
axs[1, 1].set_xlabel('Age Group')

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()
```




```
from pyspark.sql import SparkSession
from pyspark.sql.functions import mean as _mean
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Assuming df1 is your Spark DataFrame
# Grouping data by gender and calculating mean values for the
specified columns
gender_data = df1.groupBy('Gender').agg(
    _mean('Curricular units 1st sem (credited)').alias('Credited'),
    _mean('Curricular units 1st sem (enrolled)').alias('Enrolled'),
    _mean('Curricular units 1st sem (approved)').alias('Approved')
)

# Convert to Pandas DataFrame for easier manipulation and
visualization
gender_data_pd = gender_data.toPandas()

# Renaming the 'Gender' values for better visualization
gender_data_pd['Gender'] = gender_data_pd['Gender'].replace({'F':
'Female', 'M': 'Male'})

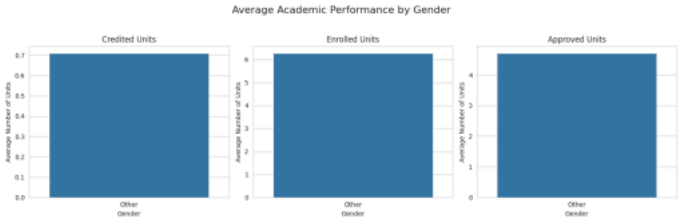
# Plotting the average number of curricular units credited, enrolled,
and approved by gender
fig, axs = plt.subplots(1, 3, figsize=(15, 5))
```

```
fig.suptitle('Average Academic Performance by Gender', fontsize=16)

# Columns to be visualized
columns_to_visualize = ['Credited', 'Enrolled', 'Approved']
titles = ['Credited Units', 'Enrolled Units', 'Approved Units']

# Creating the bar plots
for i, col in enumerate(columns_to_visualize):
    sns.barplot(x='Gender', y=col, data=gender_data_pd, ax=axs[i])
    axs[i].set_title(titles[i])
    axs[i].set_ylabel('Average Number of Units')
    axs[i].set_xlabel('Gender')

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()
```




```
from pyspark.sql.functions import mean as _mean
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Assuming df1 is your Spark DataFrame
# Grouping data by gender and calculating mean values for the
specified columns
gender_data = df1.groupBy('Gender').agg(
    _mean('Curricular units 1st sem (credited)').alias('Credited'),
    _mean('Curricular units 1st sem (enrolled)').alias('Enrolled'),
    _mean('Curricular units 1st sem (approved)').alias('Approved')
)

# Convert to Pandas DataFrame for easier manipulation and
visualization
gender_data_pd = gender_data.toPandas()

# Renaming the 'Gender' values for better visualization
gender_data_pd['Gender'] = gender_data_pd['Gender'].replace({1:
```

```
'Male', 2: 'Female', 0: 'Other'})

# Set the overall style of the plots
sns.set(style="whitegrid")

# Create a figure with 1 row and 3 columns for the subplots
fig, axs = plt.subplots(1, 3, figsize=(18, 6))
fig.suptitle('Average Academic Performance by Gender (1st Semester)',
fontsize=18, y=1.03)

# Define the columns to visualize and titles
columns_to_visualize = ['Credited', 'Enrolled', 'Approved']
titles = ['Average Credited Units', 'Average Enrolled Units', 'Average
Approved Units']

# Color palette
palette = sns.color_palette("Set2")

# Loop through the columns and plot them
for i, col in enumerate(columns_to_visualize):
    sns.barplot(x='Gender', y=col, data=gender_data_pd, ax=axs[i],
palette=palette)
    axs[i].set_title(titles[i], fontsize=14)
    axs[i].set_ylabel('Average Units', fontsize=12)
    axs[i].set_xlabel('Gender', fontsize=12)

    # Adding value annotations on the bars
    for p in axs[i].patches:
        axs[i].annotate(f'{p.get_height():.2f}',
                        (p.get_x() + p.get_width() / 2.,
p.get_height()),
                        ha='center', va='baseline',
                        fontsize=11, color='black', xytext=(0, 5),
                        textcoords='offset points')

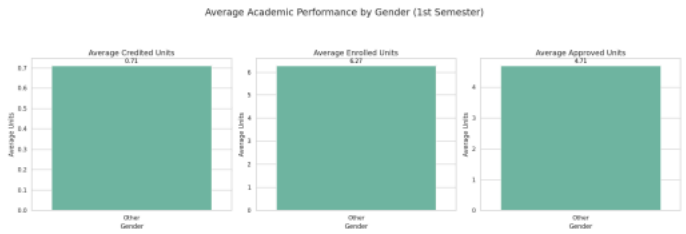
# Adjust the layout to prevent overlap
plt.tight_layout(rect=[0, 0.03, 1, 0.95])

# Display the plot
plt.show()
```

```
<ipython-input-48-f5a1ba6f4fd4>:36: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x='Gender', y=col, data=gender_data_pd, ax=axs[i],
palette=palette)
<ipython-input-48-f5a1ba6f4fd4>:36: UserWarning: The palette list has
```




```
from pyspark.sql.functions import when

# Encoding the target variable using PySpark's when function
df1 = df1.withColumn("Target_encoded", when(df1["Target"] ==
"Dropout", 1).otherwise(0))

from pyspark.sql.functions import col, isnan, when, count
```

```
# Checking for missing values in each column
missing_counts = df1.select([count(when(isnan(c) | col(c).isNull(),
c)).alias(c) for c in df1.columns])
missing_counts.show()

# Total count of missing values in the entire DataFrame
total_missing_values = sum(row[0] for row in missing_counts.collect())
print("Total missing values:", total_missing_values)
```

```
# Display the first few rows of the scaled training data (may need to
convert to DataFrame for display)
X_train_scaled[:5]
```

```
array([[-2.19189678, -1.96234356, -0.17355626, -0.46134859,
0.88478423,
         1.        ,  0.        ,  0.        ,  0.        ,  1.
,
         0.        ,  0.        ],
       [ 0.21457761,  0.21214483, -0.17355626, -0.46134859,
0.88478423,
         1.        ,  0.        ,  1.        ,  0.        ,  0.
,
         0.        ,  0.        ],
       [ 0.73025069,  0.65183918, -0.28596761,  0.11873318,
```

```
0.76159704,
         1.        ,  1.        ,  0.        ,  0.        ,  0.
,
         0.        ,  0.        ],
       [-2.19189678, -1.96234356,  1.47514349,  1.1338763 , -
1.79013779,
         1.        ,  0.        ,  0.        ,  0.        ,  0.
,
         0.        ,  1.        ],
       [ 0.67524556,  0.7046025 , -1.48502197,  0.98885585,
0.13686196,
         1.        ,  1.        ,  0.        ,  0.        ,  0.
,
         0.        ,  0.        ]])
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
# Initializing the models
rf_model = RandomForestClassifier(random_state=42)
lr_model = LogisticRegression(random_state=42)
svm_model = SVC(random_state=42)
# Training the models
rf_model.fit(X_train_scaled, y_train)
lr_model.fit(X_train_scaled, y_train)
svm_model.fit(X_train_scaled, y_train)
# Getting predictions on the test set
rf_predictions = rf_model.predict(X_test_scaled)
lr_predictions = lr_model.predict(X_test_scaled)
svm_predictions = svm_model.predict(X_test_scaled)
# Storing the models and predictions in dictionaries for easier access
models = {'Random Forest': rf_model, 'Logistic Regression': lr_model,
'Support Vector Machine': svm_model}
predictions = {'Random Forest': rf_predictions, 'Logistic Regression':
lr_predictions, 'Support Vector Machine': svm_predictions}
# Displaying the accuracy of each model
accuracy_scores = {model: accuracy_score(y_test, pred) for model, pred
in predictions.items()}
accuracy_scores
```

```
<ipython-input-58-588963d2c8e5>:10: DataConversionWarning: A column-
vector y was passed when a 1d array was expected. Please change the
shape of y to (n_samples,), for example using ravel().
  rf_model.fit(X_train_scaled, y_train)
/usr/local/lib/python3.10/dist-packages/sklearn/utils/validation.py:11
43: DataConversionWarning: A column-vector y was passed when a 1d
array was expected. Please change the shape of y to (n_samples, ), for
example using ravel().
```

```
  y = column_or_1d(y, warn=True)
/usr/local/lib/python3.10/dist-packages/sklearn/utils/validation.py:11
43: DataConversionWarning: A column-vector y was passed when a 1d
array was expected. Please change the shape of y to (n_samples, ), for
example using ravel().
  y = column_or_1d(y, warn=True)

{'Random Forest': 0.6225988700564972,
 'Logistic Regression': 0.6508474576271186,
 'Support Vector Machine': 0.6429378531073446}
```

```
from sklearn.metrics import classification_report
import pandas as pd

preds = {
    'Random Forest': rf_predictions,
    'Logistic Regression': lr_predictions,

}

# Creating classification reports for each model
classification_reports = {model: classification_report(y_test, pred,
output_dict=True)
                          for model, pred in preds.items()}

# Converting the classification reports to DataFrames for better
visualization
classification_reports_df = {model: pd.DataFrame(report).transpose()
                             for model, report in
classification_reports.items()}

# Displaying the classification reports
rf_report = classification_reports_df.get('Random Forest')
lr_report = classification_reports_df.get('Logistic Regression')

# If you want to display them one by one
print("Random Forest Classification Report:")
print(rf_report)

print("\nLogistic Regression Classification Report:")
print(lr_report)

# Optionally, display multiple reports at once using Jupyter's display
functionality
from IPython.display import display

print("\nClassification Report for Random Forest:")
display(rf_report)

print("\nClassification Report for Logistic Regression:")
display(lr_report)
```

```
Random Forest Classification Report:
              precision    recall  f1-score     support
Dropout        0.664430  0.626582  0.644951  316.000000
Enrolled       0.264151  0.185430  0.217899  151.000000
Graduate       0.675676  0.777512  0.723026  418.000000
accuracy       0.622599  0.622599  0.622599    0.622599
macro avg      0.534752  0.529842  0.528625  885.000000
weighted avg   0.601445  0.622599  0.608963  885.000000

Logistic Regression Classification Report:
              precision    recall  f1-score     support
Dropout        0.777293  0.563291  0.653211  316.000000
Enrolled       0.500000  0.006623  0.013072  151.000000
Graduate       0.607034  0.949761  0.740672  418.000000
accuracy       0.650847  0.650847  0.650847    0.650847
macro avg      0.628109  0.506558  0.468985  885.000000
weighted avg   0.649564  0.650847  0.585299  885.000000

Classification Report for Random Forest:
```


```
{"summary":"{\n  \"name\": \"rf_report\",\n  \"rows\": 6,\n
\"fields\": [\n    {\n      \"column\": \"precision\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.15367841631209622,\n        \"min\": 0.2641509433962264,\n
\"max\": 0.6756756756756757,\n        \"num_unique_values\": 6,\n
\"samples\": [\n          0.6644295302013423,\n
0.2641509433962264,\n          0.6014451485072166\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"recall\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.20039475407606902,\n        \"min\": 0.18543046357615894,\n
\"max\": 0.777511961722488,\n        \"num_unique_values\": 5,\n
\"samples\": [\n          0.18543046357615894,\n
0.5298415679265532,\n          0.777511961722488\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"f1-score\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.17781744097205351,\n        \"min\": 0.2178988326848249,\n
\"max\": 0.7230255839822024,\n        \"num_unique_values\": 6,\n
\"samples\": [\n          0.6449511400651465,\n
0.2178988326848249,\n          0.6089626871192718\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"support\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
371.0640977040115,\n        \"min\": 0.6225988700564972,\n
\"max\": 885.0,\n        \"num_unique_values\": 5,\n
\"samples\": [\n          151.0,\n          885.0,\n          418.0\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    }\n  ]\n}","type":"dataframe","variable_name":"rf_report"}
```

Classification Report for Logistic Regression:


```
{"summary":"{\n  \"name\": \"lr_report\",\n  \"rows\": 6,\n
\"fields\": [\n    {\n      \"column\": \"precision\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.0891781338049641,\n        \"min\": 0.5,\n        \"max\":
0.777292576419214,\n        \"num_unique_values\": 6,\n
\"samples\": [\n          0.777292576419214,\n          0.5,\n
0.6495644240797188\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"recall\",\n      \"properties\": {\n        \"dtype\": \"number\",\n
\"std\": 0.3089796231105855,\n        \"min\": 0.006622516556291391,\n
\"max\": 0.9497607655502392,\n        \"num_unique_values\": 5,\n
\"samples\": [\n          0.006622516556291391,\n
0.5065581404490124,\n          0.9497607655502392\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"f1-score\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.26358843779804825,\n        \"min\": 0.0130718954248366,\n
\"max\": 0.7406716417910447,\n        \"num_unique_values\": 6,\n
\"samples\": [\n          0.653211009174312,\n
0.0130718954248366,\n          0.585298623024734\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"support\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
371.05736834134177,\n        \"min\": 0.6508474576271186,\n
\"max\": 885.0,\n        \"num_unique_values\": 5,\n
\"samples\": [\n          151.0,\n          885.0,\n          418.0\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    }\n  ]\n}","type":"dataframe","variable_name":"lr_report"}
```